

Conclusion & advanced topics

CS424: Visualization & Visual Analytics
Fabio Miranda



urbantk.org



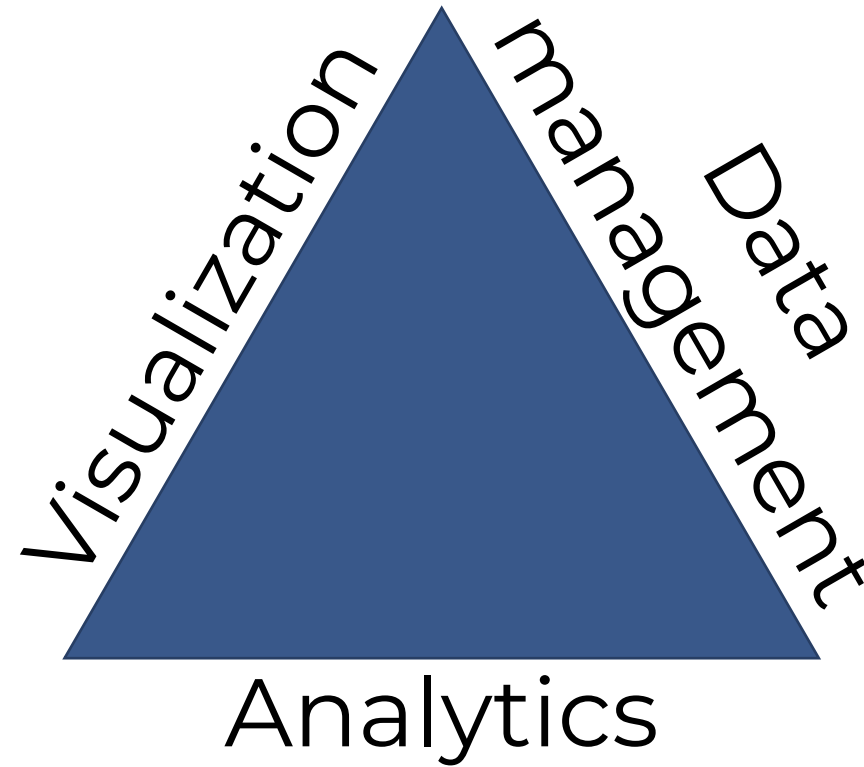
evl

electronic
visualization
laboratory

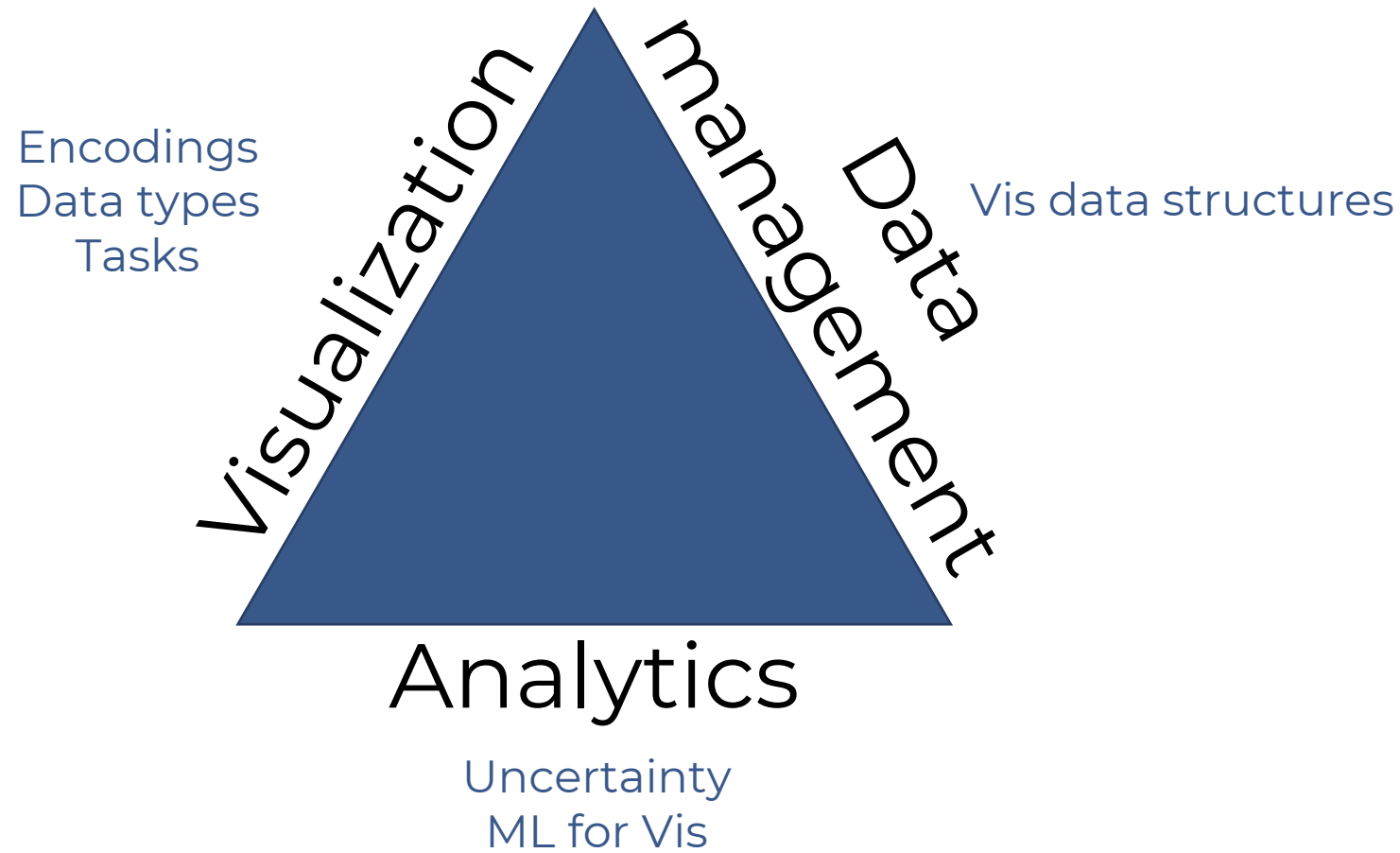


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Visual analytics

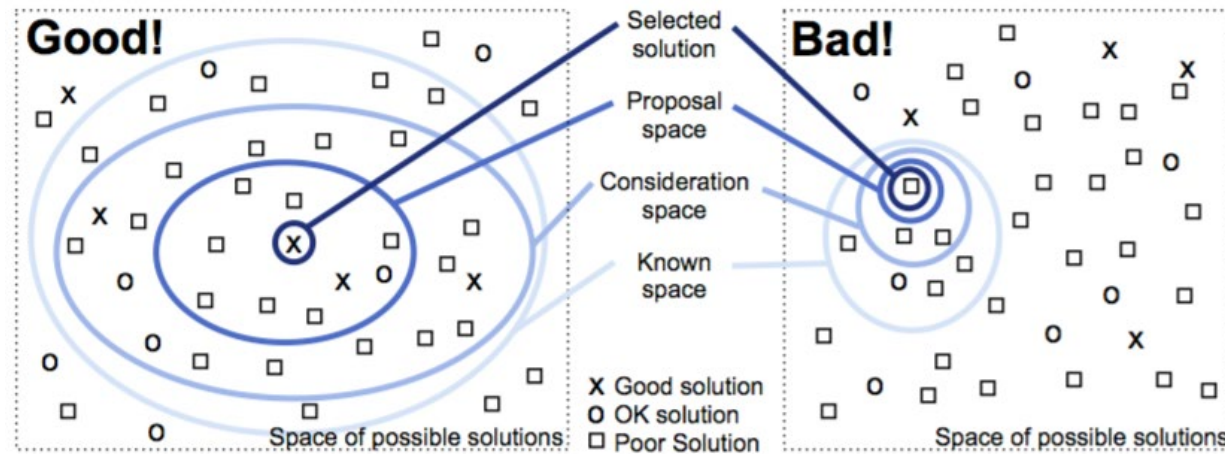


Visual analytics



Visualization design

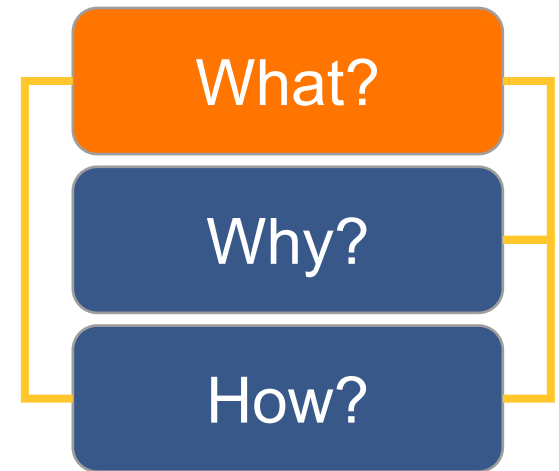
Develop principles and techniques to build effective visualizations.



[Munzner, 2015]

Visualization design

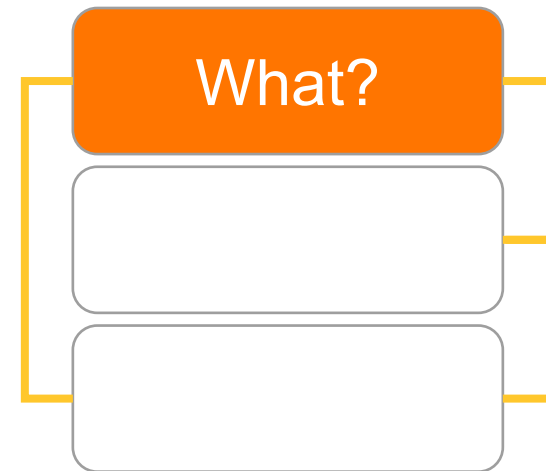
- High-level framework for analyzing vis use:
 - **What** data user sees?
 - **Why** the user intends to use a vis tool?
 - **How** the user intends to use a vis tool?



[Munzner, 2015]

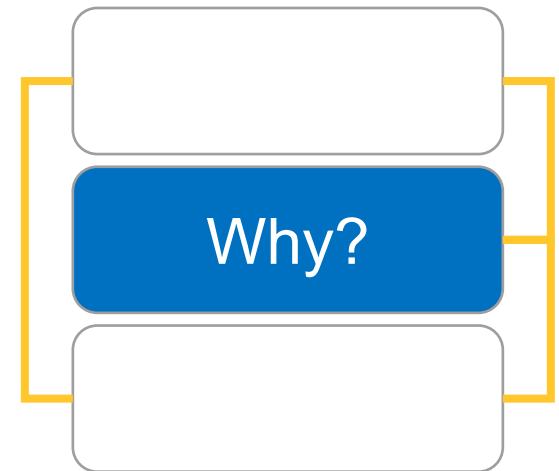
Principles of visualization

- Data
- Visual marks
- Visual channels
- Interaction



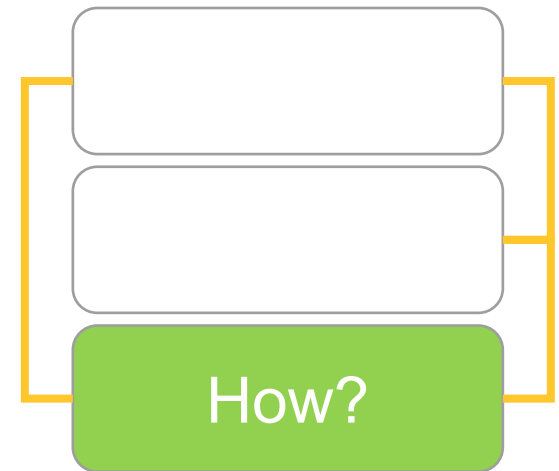
Task abstraction

- Analyzing tasks abstractly – rather than thinking of domain-specific tasks, think of abstract tasks.
- Domain-specific task: “contrast the prognosis of patients who were intubated in the ICU more than one month to patients hospitalized within the first week.”
- Abstract tasks: “compare values between two groups.”



How to design vis

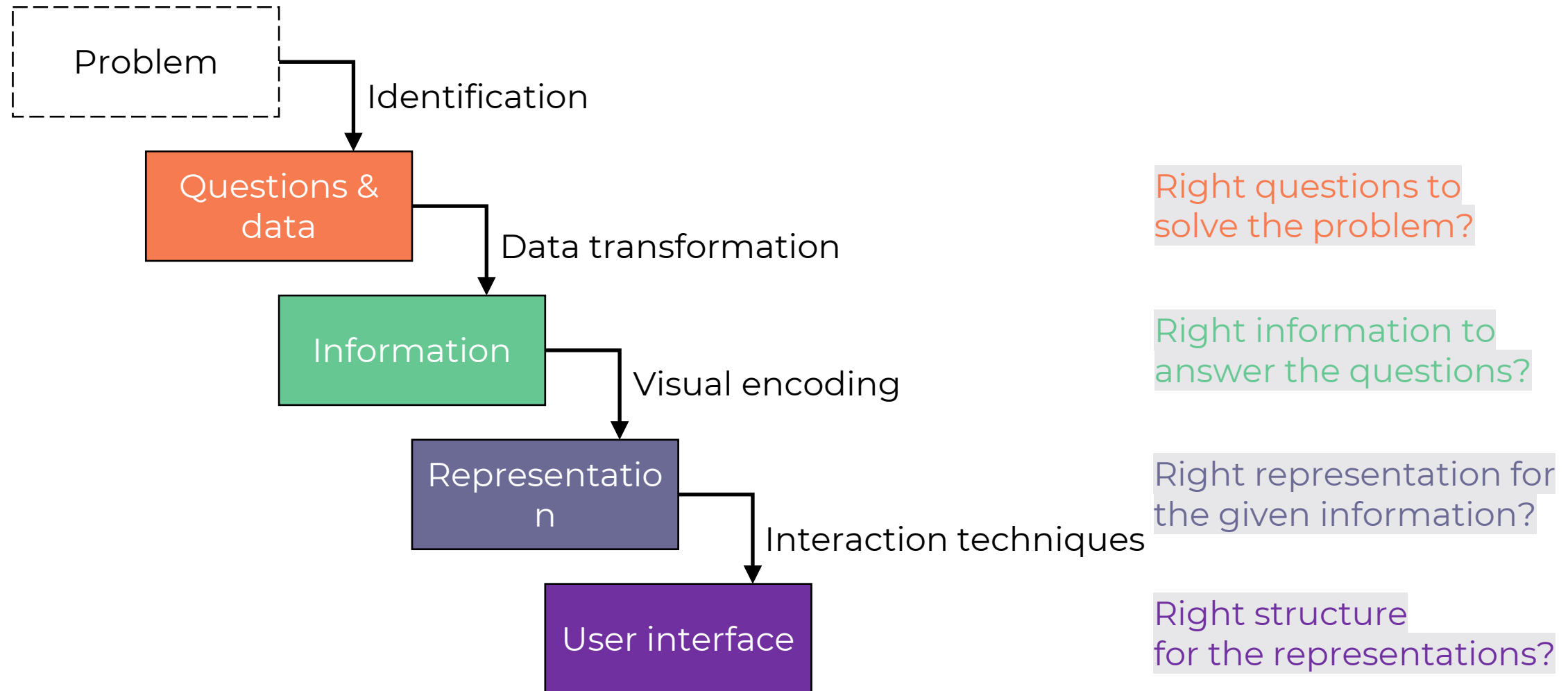
- How a vis idiom can be constructed out of a set of design choices?
 - Encode
 - Manipulate: change, select, navigate
 - Facet: coordinate multiple views
 - Reduce: filter, aggregate



How to visualize data

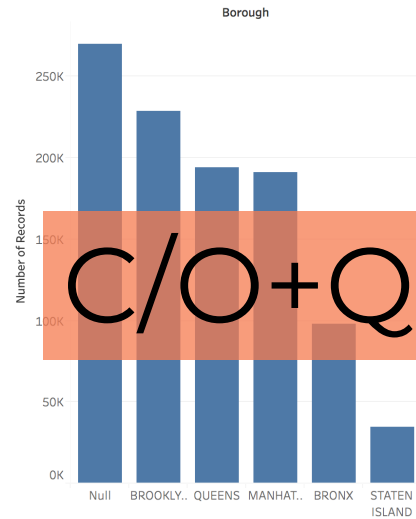
1. Decide what to visualize → select & transform data
2. Decide how to visualize it → choose or design representations

Creating visualizations

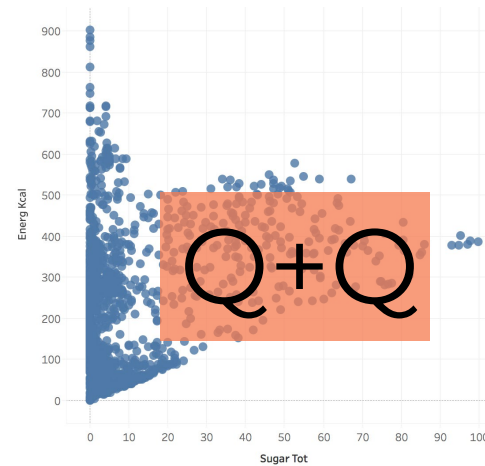


Fundamental types of visualizations

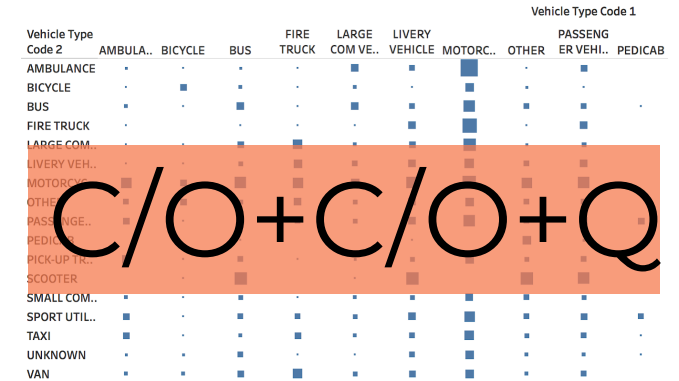
Bar chart



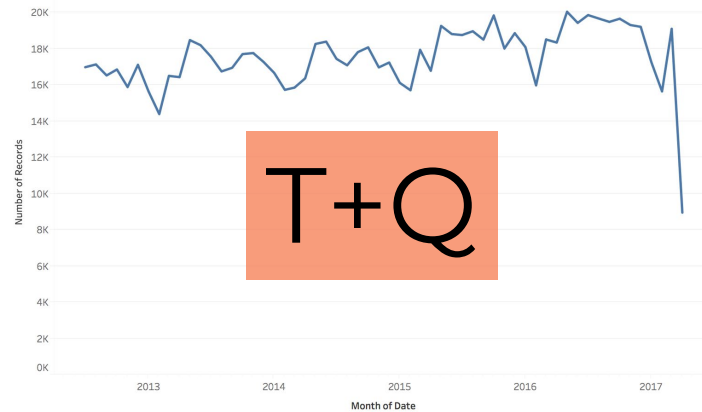
Scatter plot



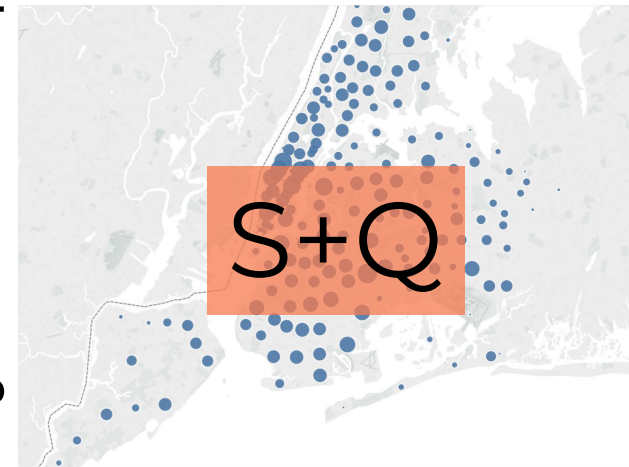
Matrix



Line chart

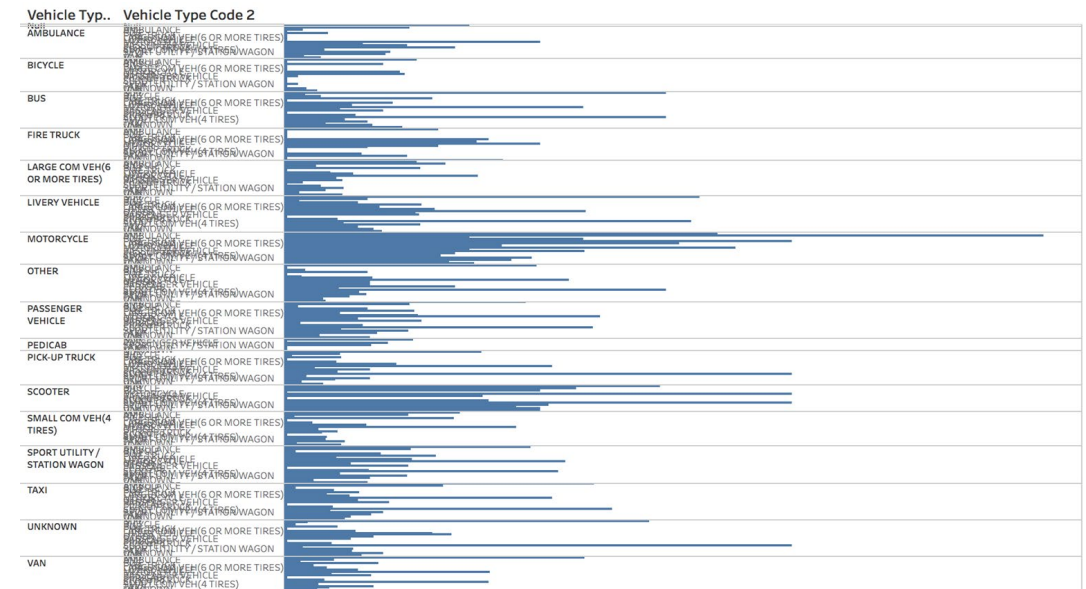
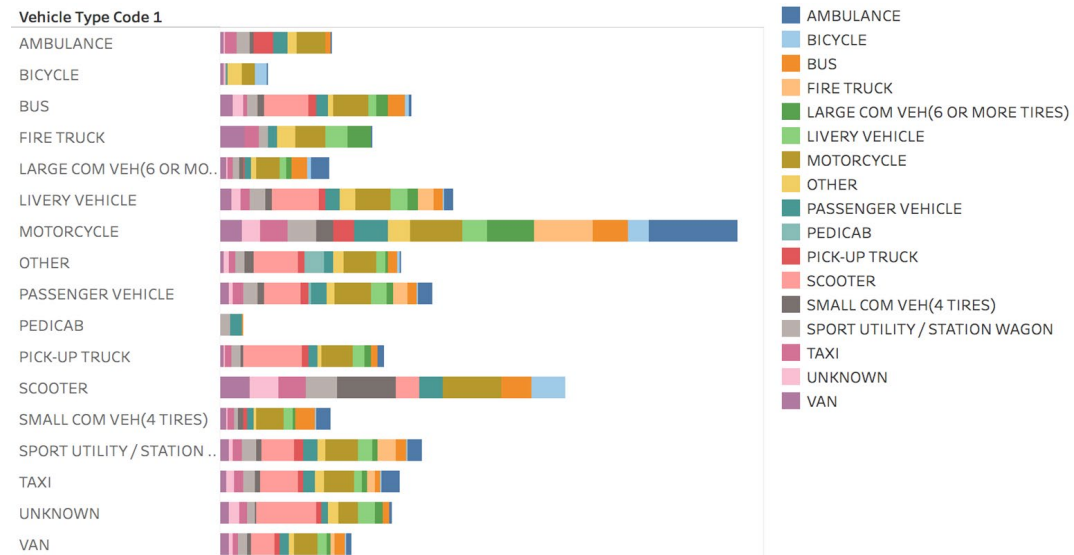


Symbol map



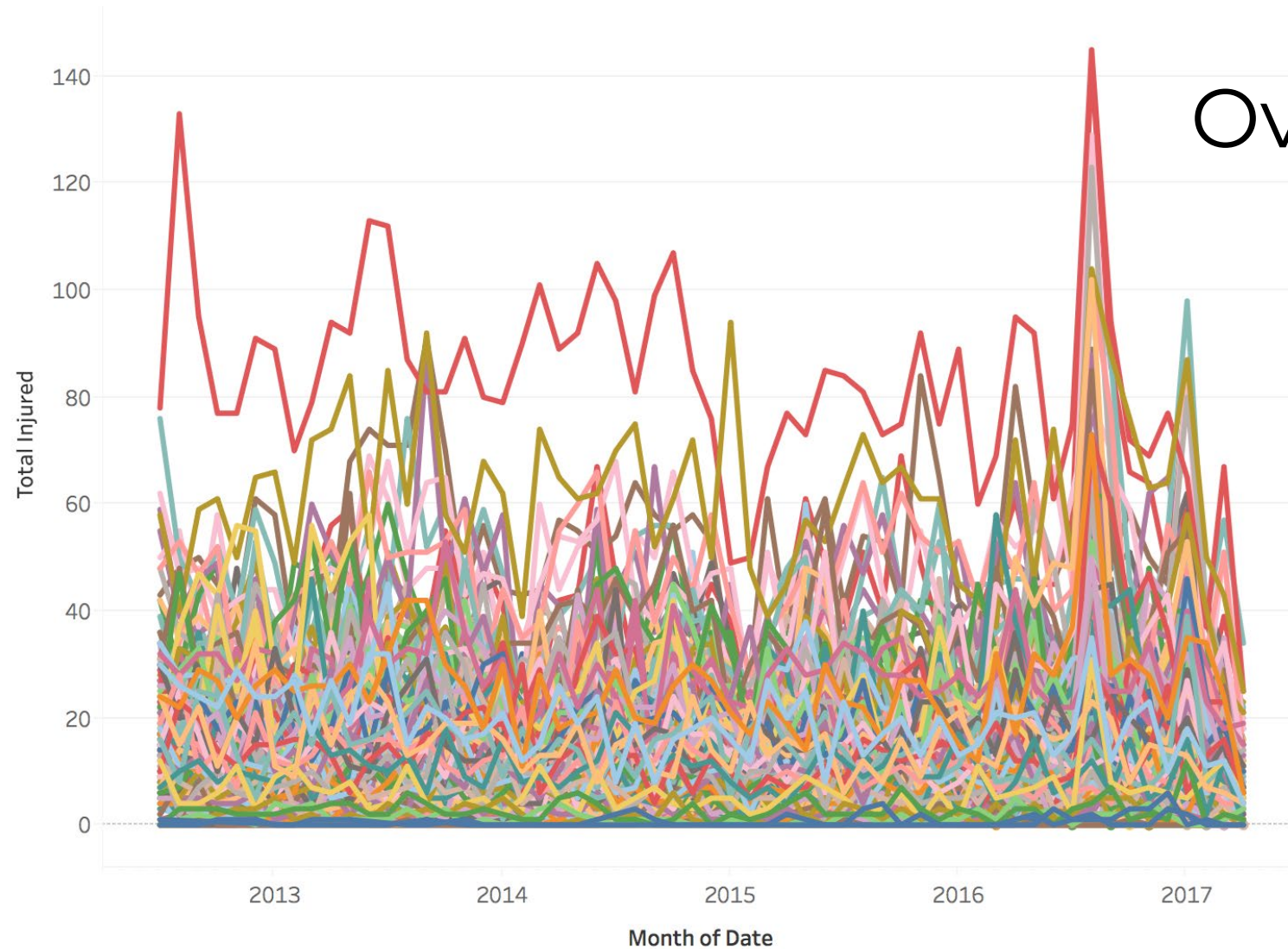
Problems

Vehicle Type Code 2	AMBULA..	BICYCLE	BUS	FIRE TRUCK	LARGE COM VE..	LIVERY VEHICLE	MOTORCYC..	OTHER	PASSENG ER VEHI..	PEDICAB
AMBULANCE	.	.	.	.	.	.	.	.	.	.
BICYCLE	.	.	.	.	.	.	.	.	.	.
BUS	.	.	.	.	.	.	.	.	.	.
FIRE TRUCK	.	.	.	.	.	.	.	.	.	.
LARGE COM..	.	.	.	.	.	.	.	.	.	.
LIVERY VEH..	.	.	.	.	.	.	.	.	.	.
MOTORCYC..	.	.	.	.	.	.	.	.	.	.
OTHER	.	.	.	.	.	.	.	.	.	.
PASSENGE..	.	.	.	.	.	.	.	.	.	.
PEDICAB	.	.	.	.	.	.	.	.	.	.
PICK-UP TR..	.	.	.	.	.	.	.	.	.	.
SCOOTER	.	.	.	.	.	.	.	.	.	.
SMALL COM..	.	.	.	.	.	.	.	.	.	.
SPORT UTIL..	.	.	.	.	.	.	.	.	.	.
TAXI	.	.	.	.	.	.	.	.	.	.
UNKNOWN	.	.	.	.	.	.	.	.	.	.
VAN	.	.	.	.	.	.	.	.	.	.

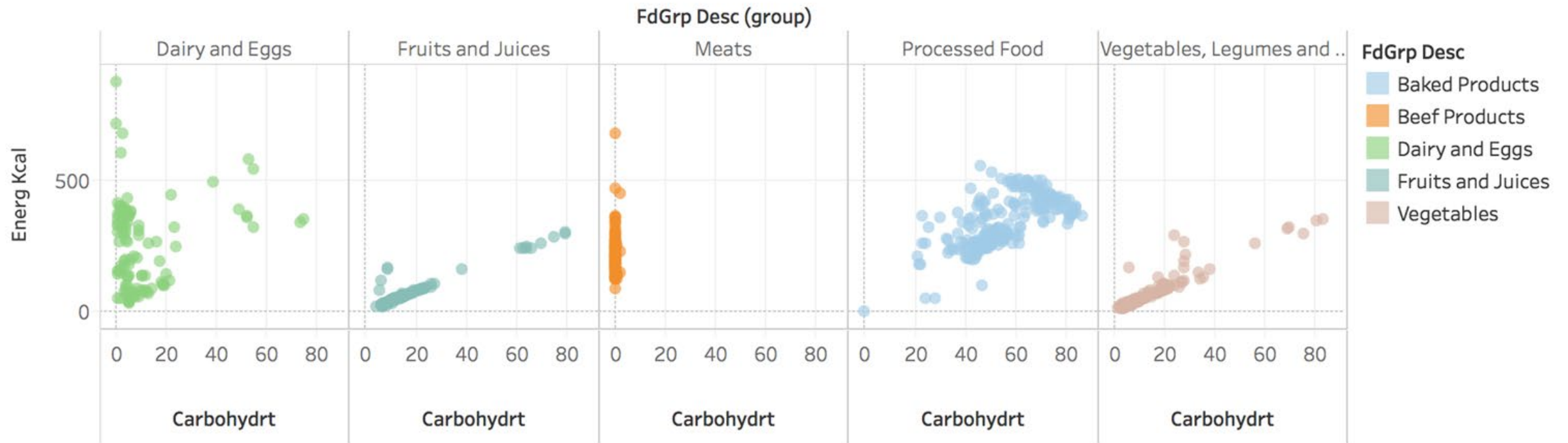


[Enrico Bertini]

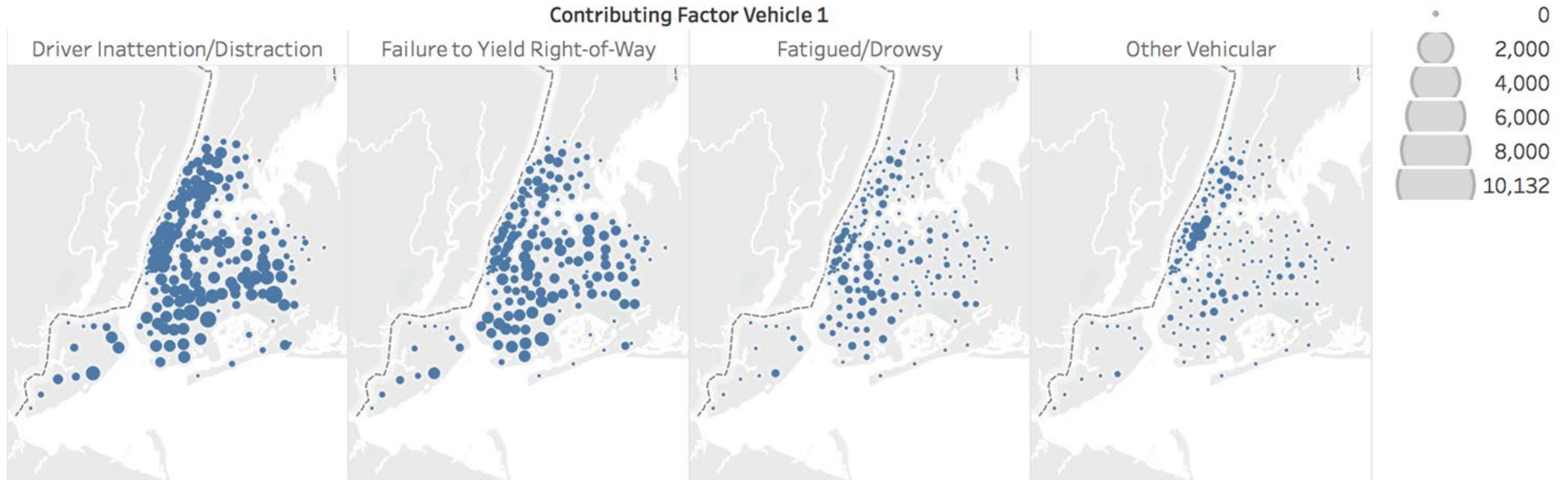
Problems



Faceting + scatter plots



Faceting + maps

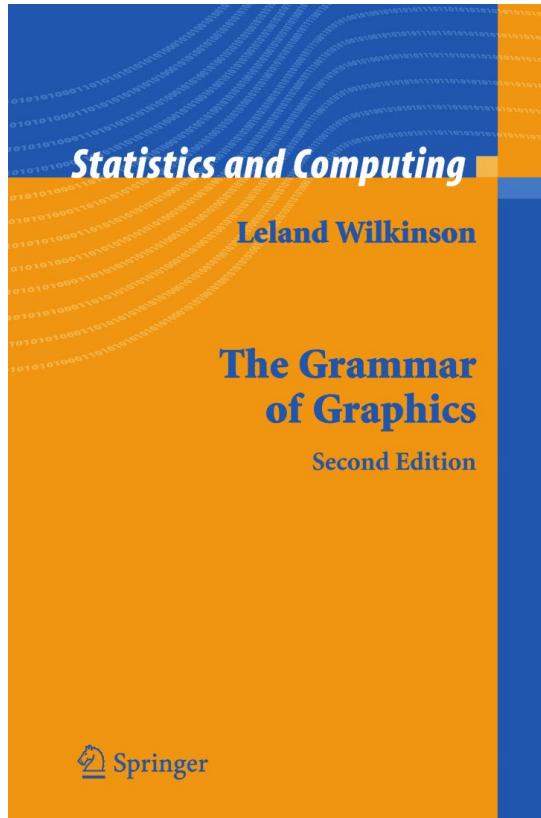


[Enrico Bertini]

Spectrum of visualization tools

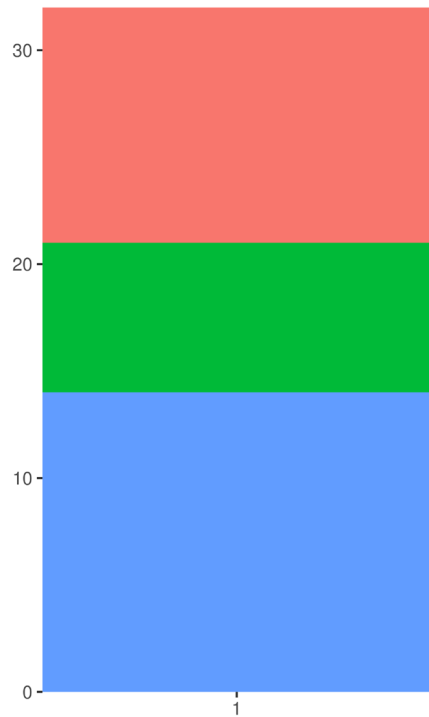


Grammar of graphics

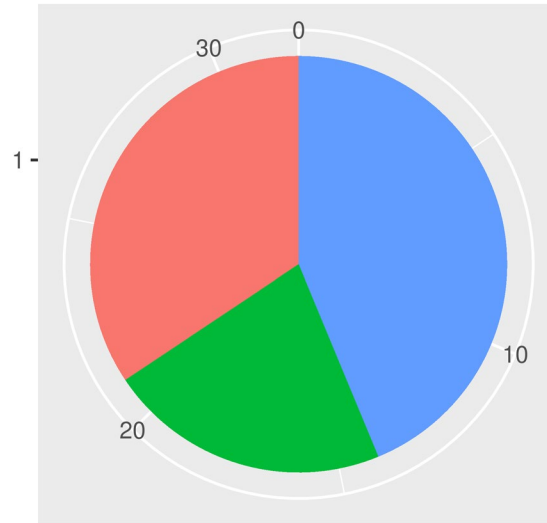


- A grammar to describe and create a wide range of statistical graphics.
- Concisely describe the components of a graphic.

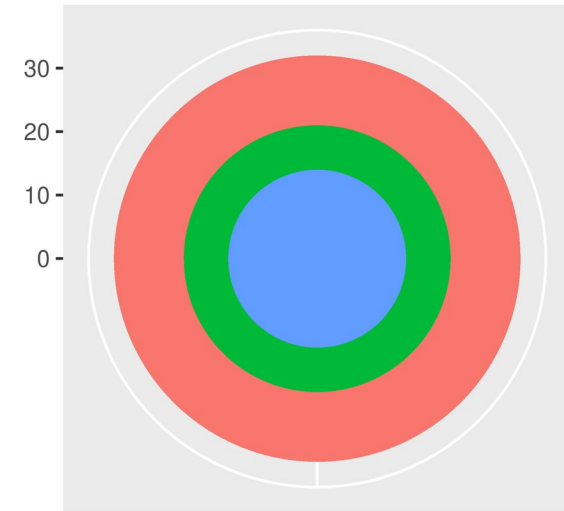
A layered grammar of graphics



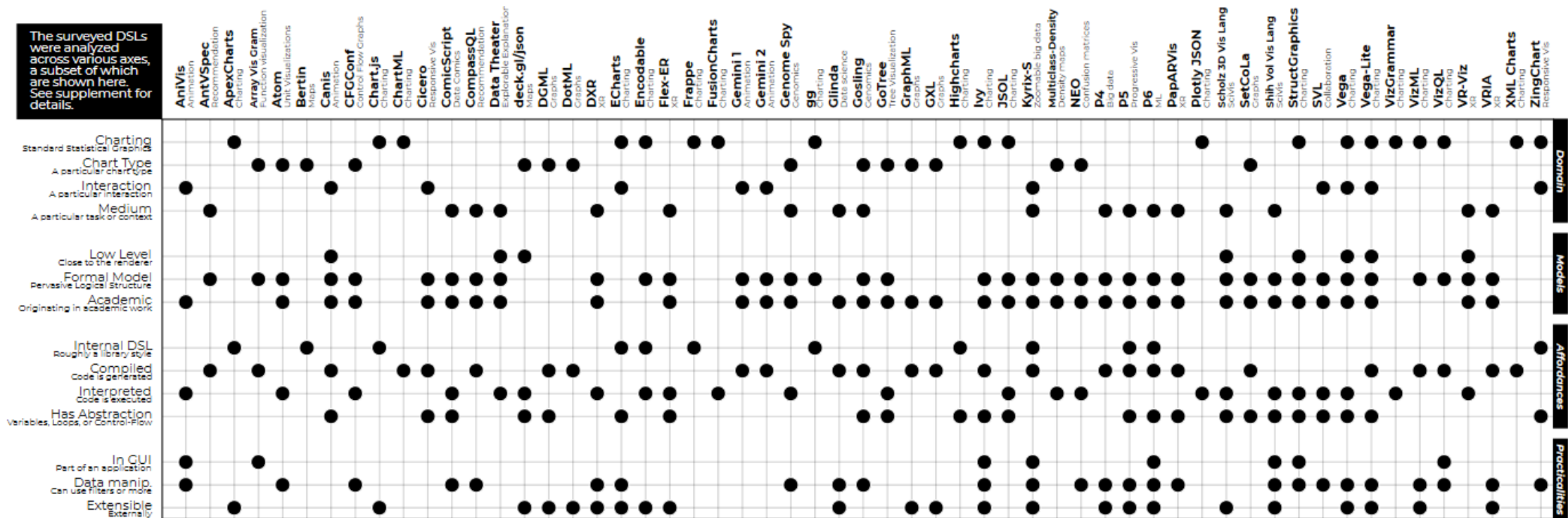
`base + coord_polar(theta = "x")`



`base + coord_polar(theta = "y")`



JSON-based grammars



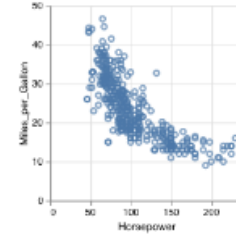
[McNutt, 2022]

JSON-based grammars

Vega-Lite

```
{data: {url: "data/cars.json"},
 mark: "point",
 encoding: {
   x: {field: "Horsepower", type: "quantitative"},
   y: {field: "Miles_per_Gallon",
      type: "quantitative"}}}
```

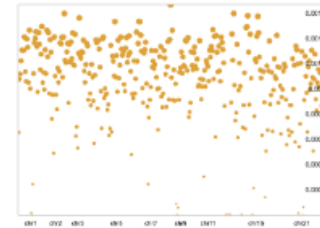
CoG-style mappings (as here) provide a fast-flexible control over the chart design space allowing smooth exploration and clear documentation of intent.



Gosling

```
{tracks: [{layout: "linear", width: 600, height: 400,
 data: {url: "<URL>", type: "multivec",
   row: "sample", column: "position",
   value: "peak", categories: ["sample 1"]},
 mark: "point",
 x: {field: "position", type: "genomic", axis: "bottom"},
 y: {field: "peak", type: "quantitative", axis: "right"},
 size: {field: "peak", type: "quantitative"}}]}
```

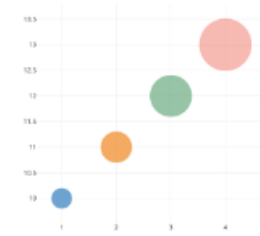
CoG-style languages can be augmented with domain specific content and abstractions, such as in the notion of tracks and genomic coordinates shown here.



Plotly JSON

```
[{x: [1, 2, 3, 4], y: [10, 11, 12, 13], mode: "markers",
 marker: {
   color: ["rgb(93, 164, 214)", "rgb(255, 144, 14)",
     "rgb(44, 160, 101)", "rgb(255, 65, 54)"],
   opacity: [1, 0.8, 0.6, 0.4],
   size: [40, 60, 80, 100]}]}
```

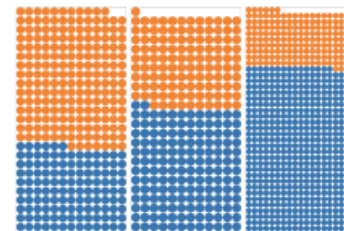
Properties (such as color) in series-models can be applied directly to the relevant component, giving a good *closeness of mapping*.



Atom

```
{data: {url: "data/titanic3.csv"},
 layouts: [
   {type: "gridxy", aspect_ratio: "fillX",
    subgroup: {type: "groupby", key: "pclass"}},
   {type: "gridxy", aspect_ratio: "maxfill",
    subgroup: {type: "flatten"},
    size: {type: "uniform", isShared: false}},
   mark: {shape: "circle", color: {key: "survived"}}}
```

Alternative formal models (such as the L-System inspired framing used here) can motivate alternative analyses and creation of novel chart forms.



[McNutt, 2022]

Vega-Lite specification

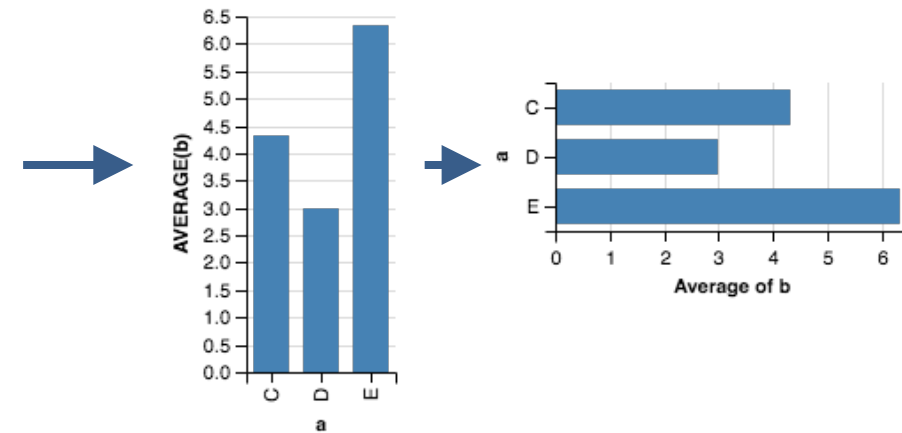
- **Data** source
- Data **transformation** (optional)
- Graphical **mark** type
- **Encoding** / mapping between data and encoding channels (x, y, color, etc.)
- Faceting

```
{  
  "data": {  
    ...  
  },  
  "mark": ...,  
  "encoding": {  
    ...  
  }  
}
```



- Vega-Lite specification

```
{
  "data": {
    "values": [
      {"a": "C", "b": 2}, {"a": "C", "b": 7}, {"a": "C", "b": 4},
      {"a": "D", "b": 1}, {"a": "D", "b": 2}, {"a": "D", "b": 6},
      {"a": "E", "b": 8}, {"a": "E", "b": 4}, {"a": "E", "b": 7}
    ]
  },
  "mark": "point",
  "encoding": {
    "y": {"field": "a", "type": "nominal"},
    "x": {"aggregate": "mean", "field": "b", "type": "quantitative"}
  }
}
```



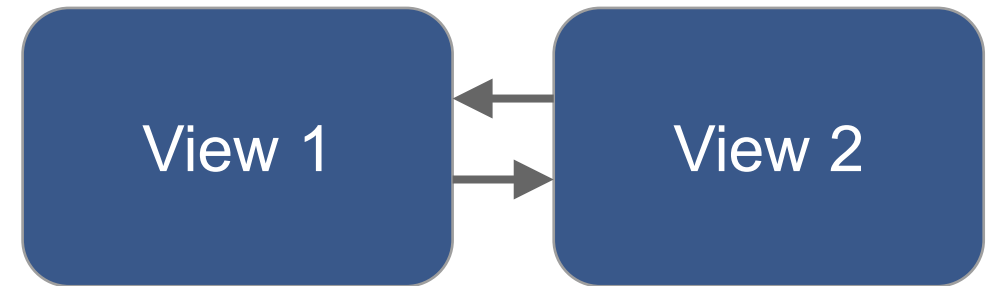
Transposing the bar chart.

Single view interactions

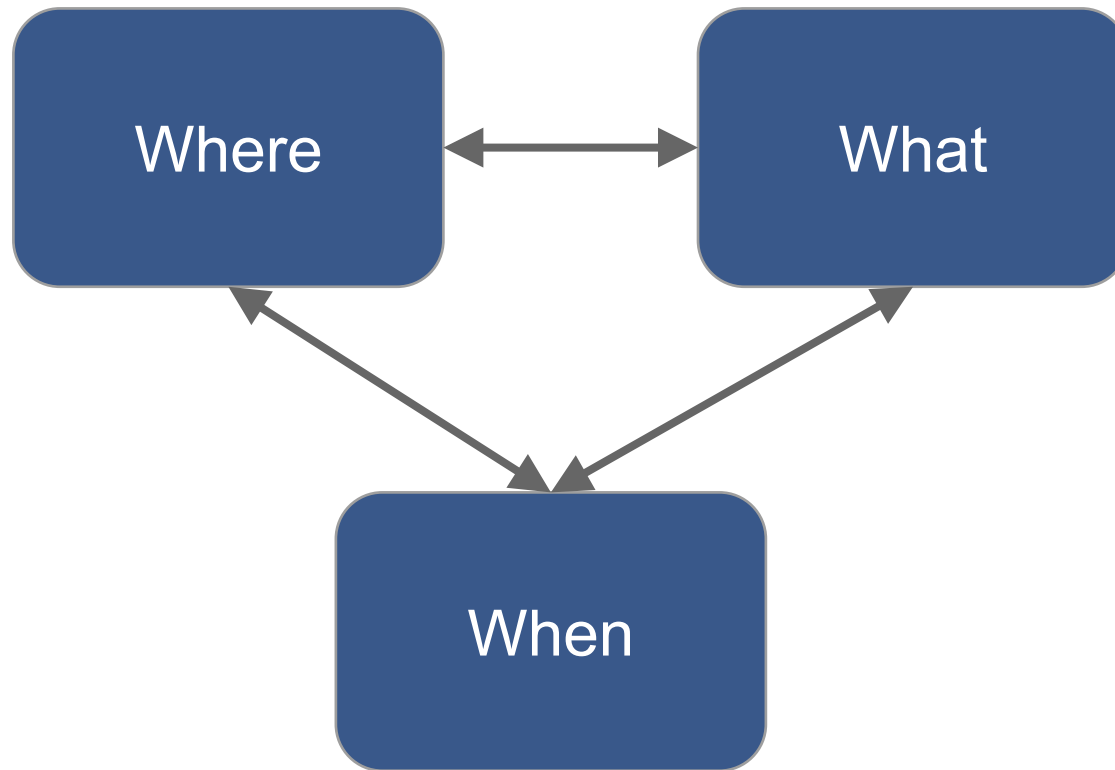
Manipulate	Methods
Data	Aggregation Filtering
Mapping	Change mapping
View	Selection Navigation Spatial arrangement



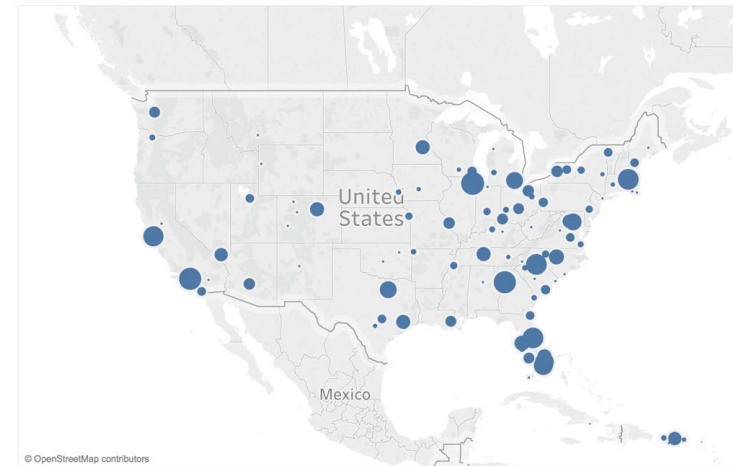
- Multiple linked views
 - How to show different properties?
 - Different information
 - Subset of data
 - Different attributes
 - Different granularity
 - Transformation
 - Different representation



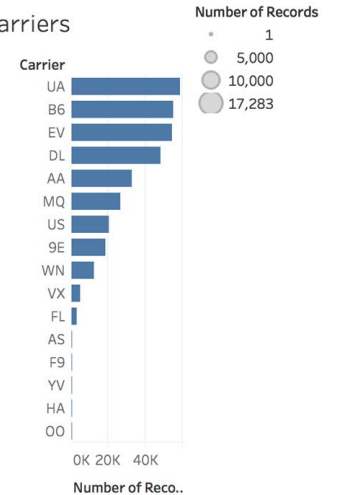
Where, what, when



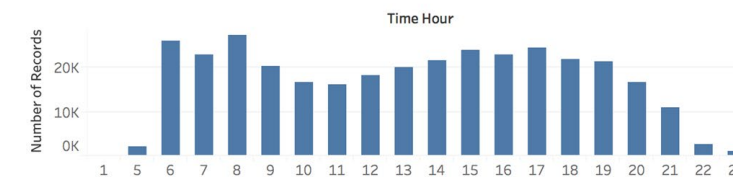
Destinations



Carriers



Departure Time

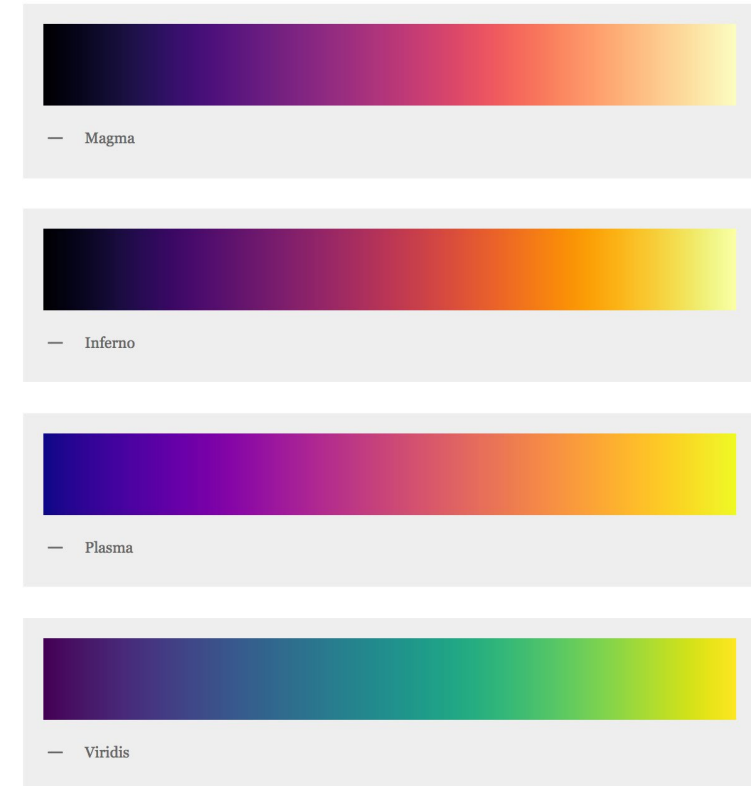


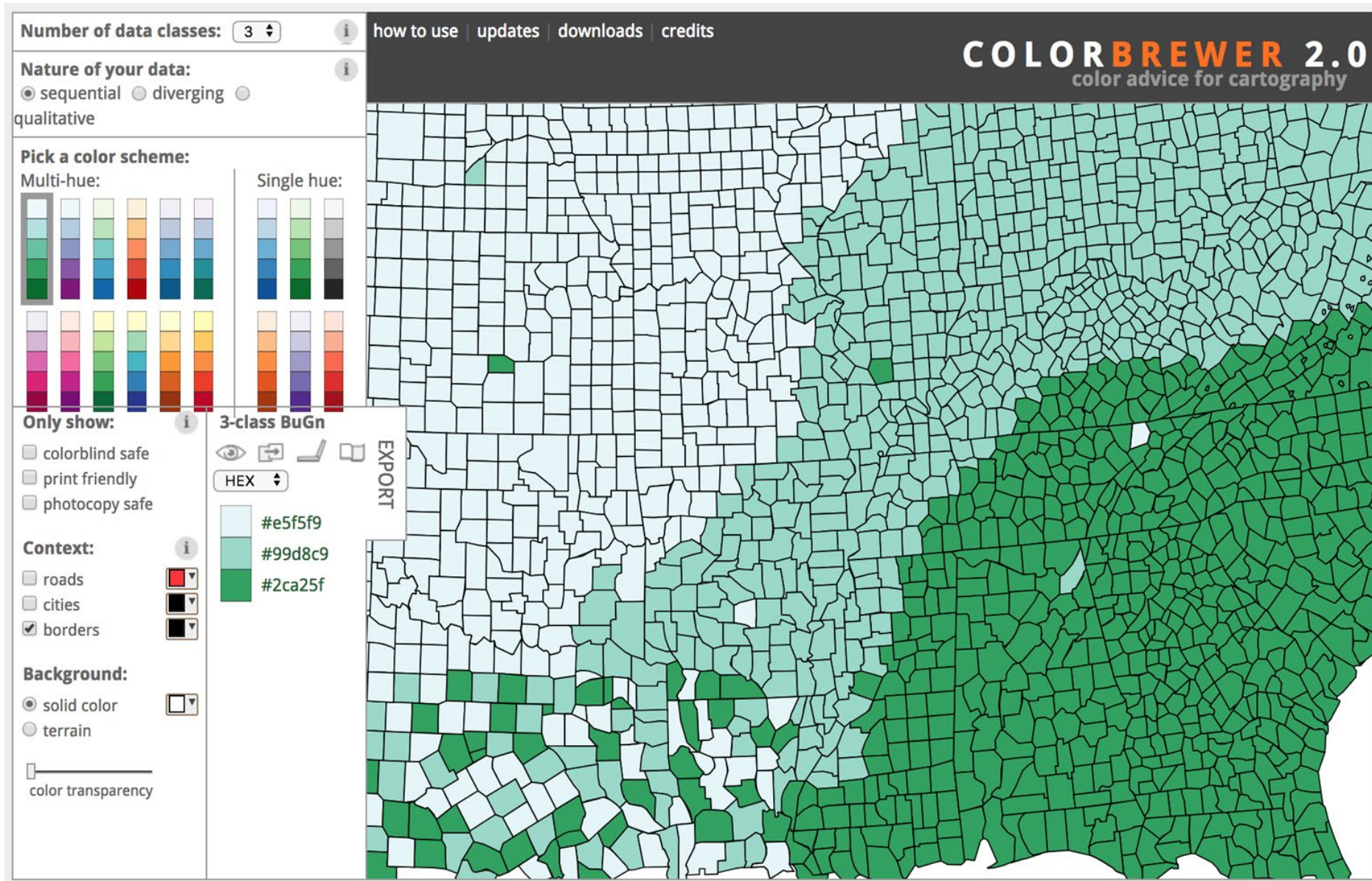
Choice of visualization

- Parameters for choosing best visualization:
 - Distribution of **items** or **values**.
 - Distribution is discrete or continuous.
1. Dot maps (distribution of **items**)
 2. Heat maps (distribution of **values**, continuous)
 3. Binned maps (distribution of **values**, continuous)
 4. Choropleth maps (distribution of **values**, discrete)
 5. Symbol maps (distribution of **values**, discrete)

Color scales

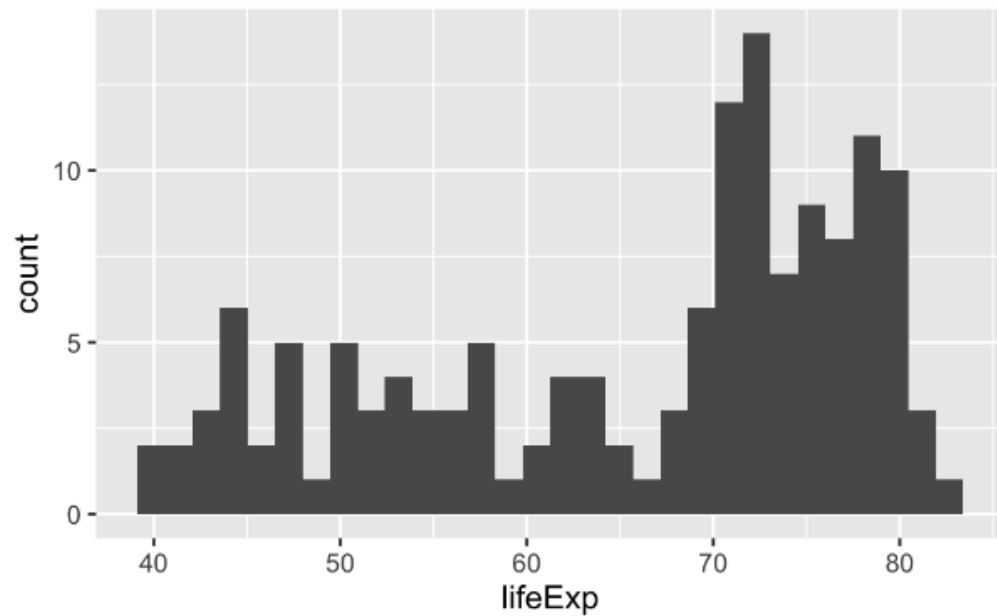
- Perceptually uniform color scales should be used instead:
- Equal steps in data values should map to equal steps in perceived color.



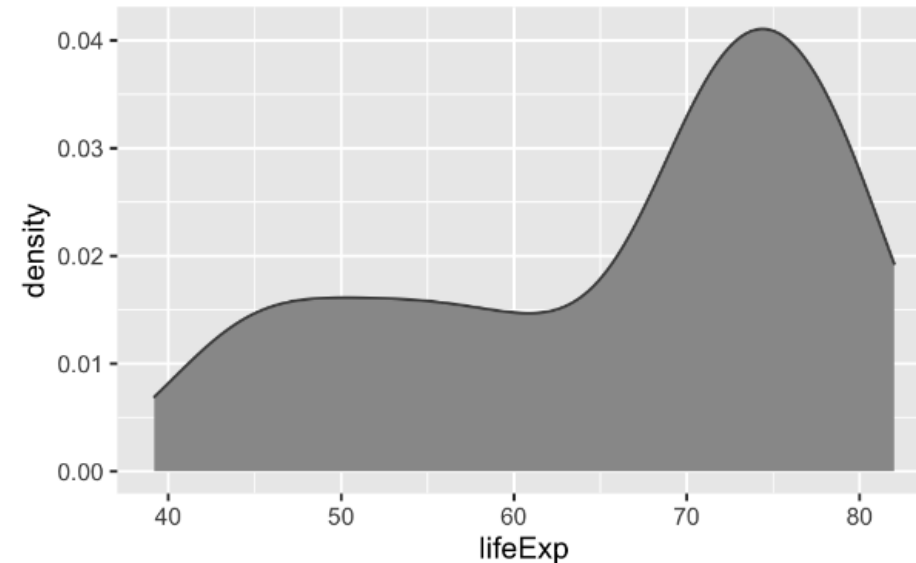


Distributions

- Histogram: approximate representation of the distribution of the data.

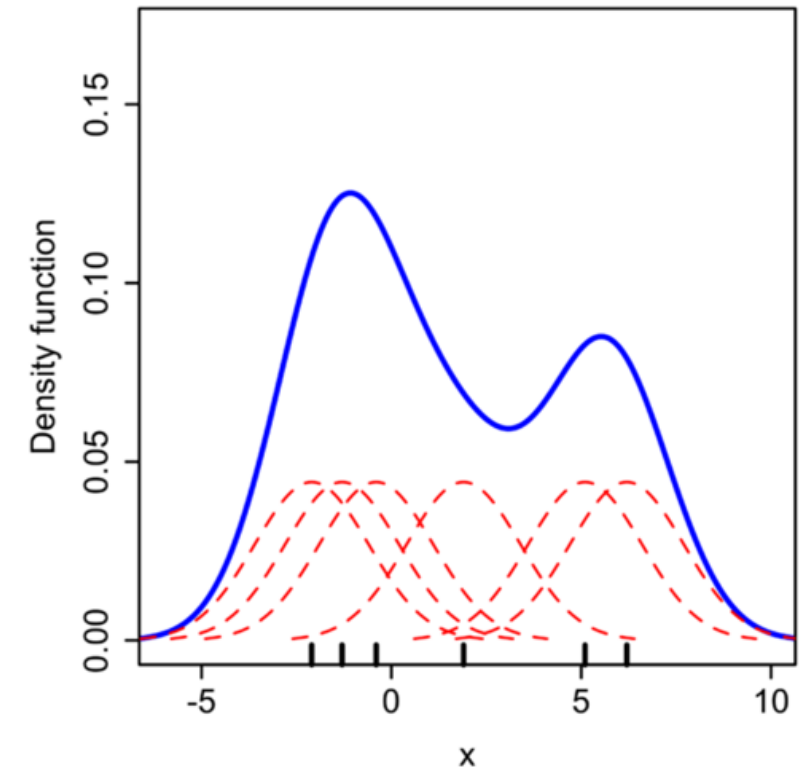


Density plots: distribution of data over continuous interval.



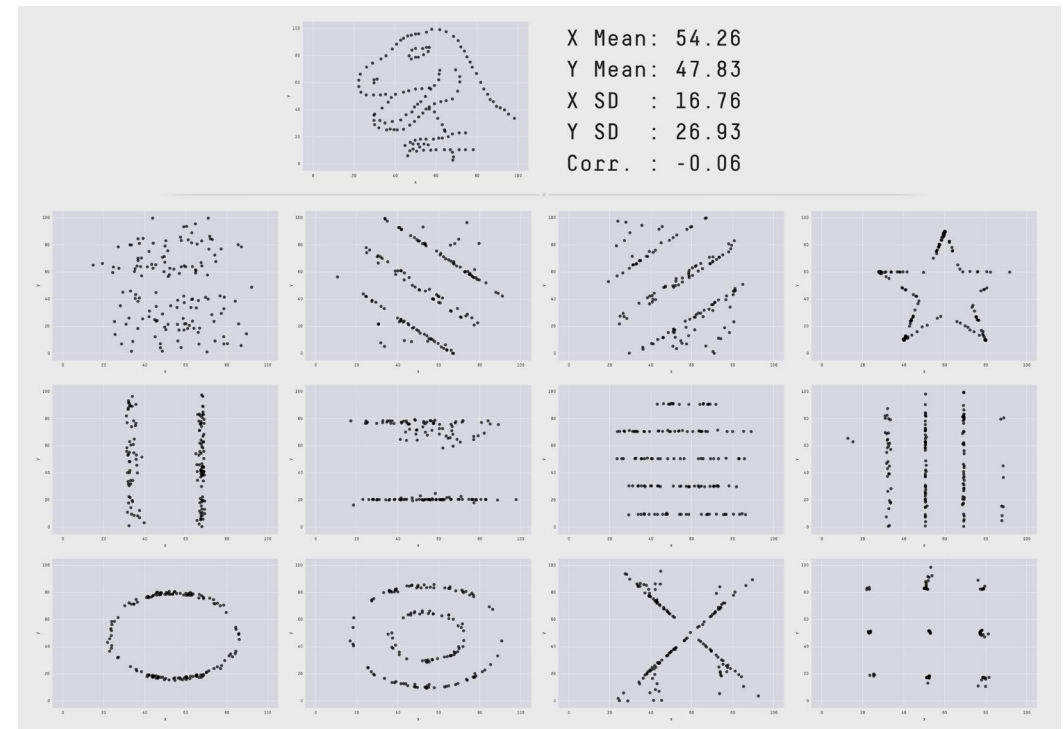
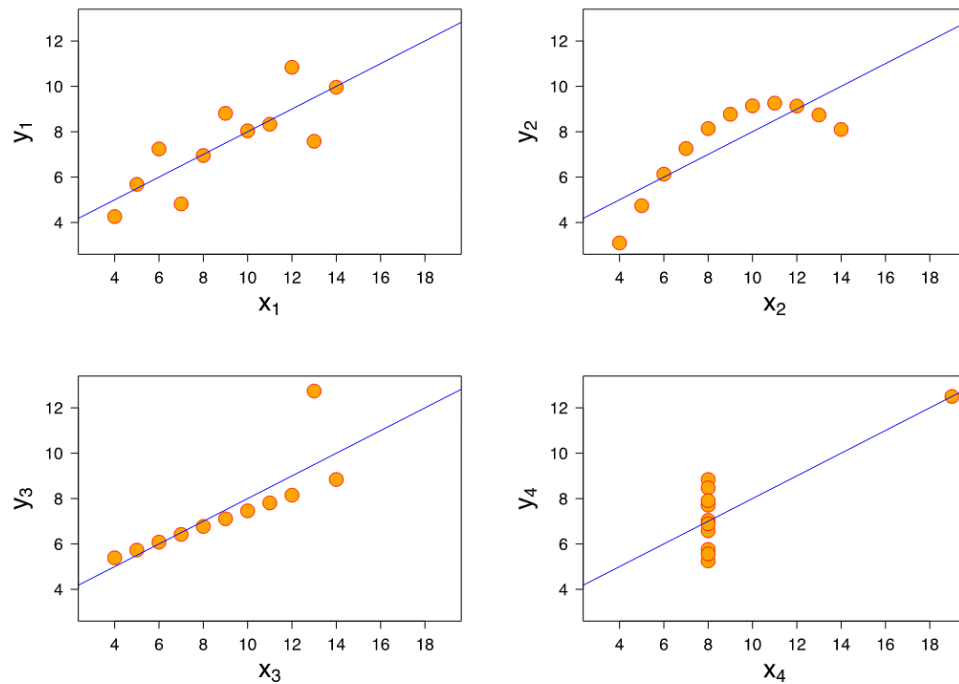
Kernel density estimation

- Nonparametric method to smooth probabilities across the range of outcomes for a random variable.
- Centers a function (i.e., kernel) at each data point and sums them to get a density estimation.



Data aggregation generates uncertainty

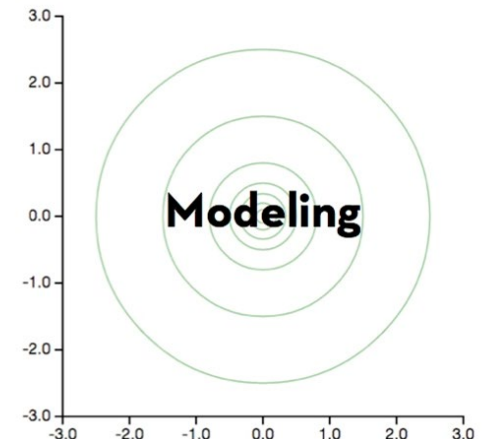
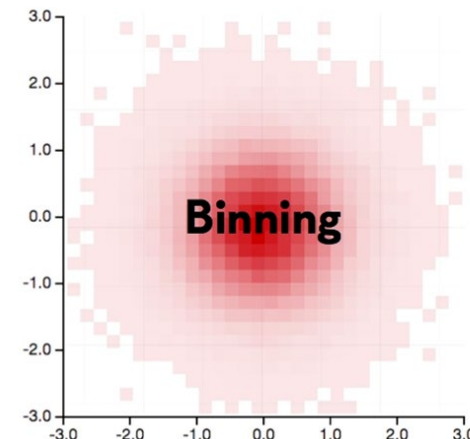
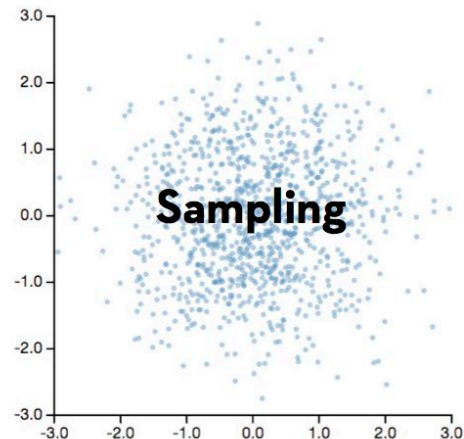
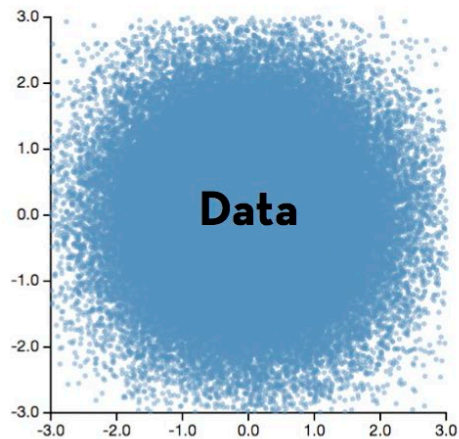
- Same stats, different graphs [Matejka and Fitzmaurice, 2017]



Summary statistics

- Summarizes statistical information
- Measures of locality:
 - Mean
 - Median
 - Interquartile mean
- Measure of spread:
 - Standard deviation
 - Range
 - Variance
 - Absolute deviation
 - Interquartile range

Data reduction techniques



Different reduction techniques applied on the same dataset

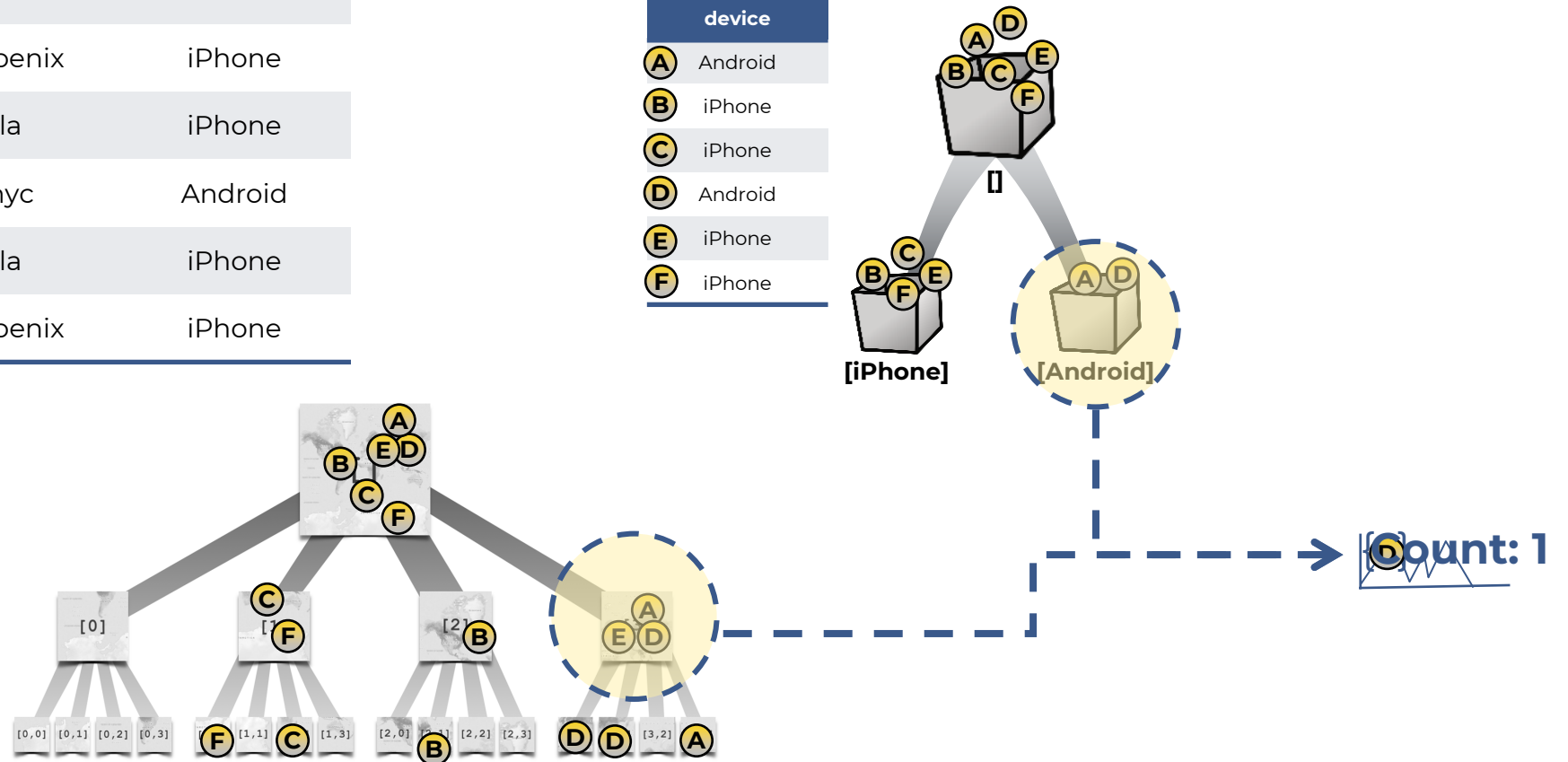
[Liu et al., 2013]

Nanocubes hierarchy

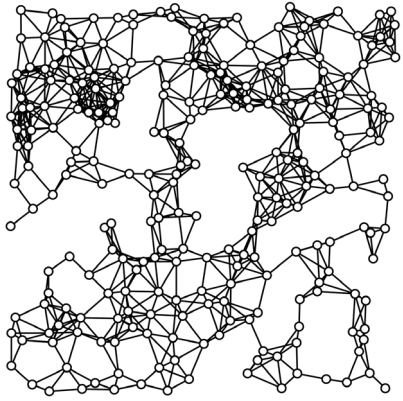
latitude	longitude	keyword	device
42.102908	-73.242852	#phoenix	Android
29.617161	-81.636398	#phoenix	iPhone
23.014051	75.120052	#la	iPhone
26.014051	75.120052	#nyc	Android
28.014051	74.120052	#la	iPhone
23.014051	75.120052	#phoenix	iPhone

	latitude	longitude
A	42.102908	-73.242852
B	29.617161	-81.636398
C	23.014051	75.120052
D	26.014051	75.120052
E	28.014051	74.120052
F	29.61161	-81.63638

device
A Android
B iPhone
C iPhone
D Android
E iPhone
F iPhone



What are embeddings?



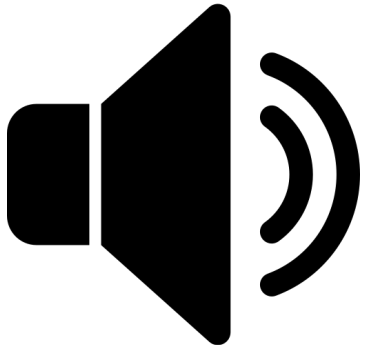
Embedding
model

0.5 0.3 0.1 0.8 ... 0.9 0.1



Embedding
model

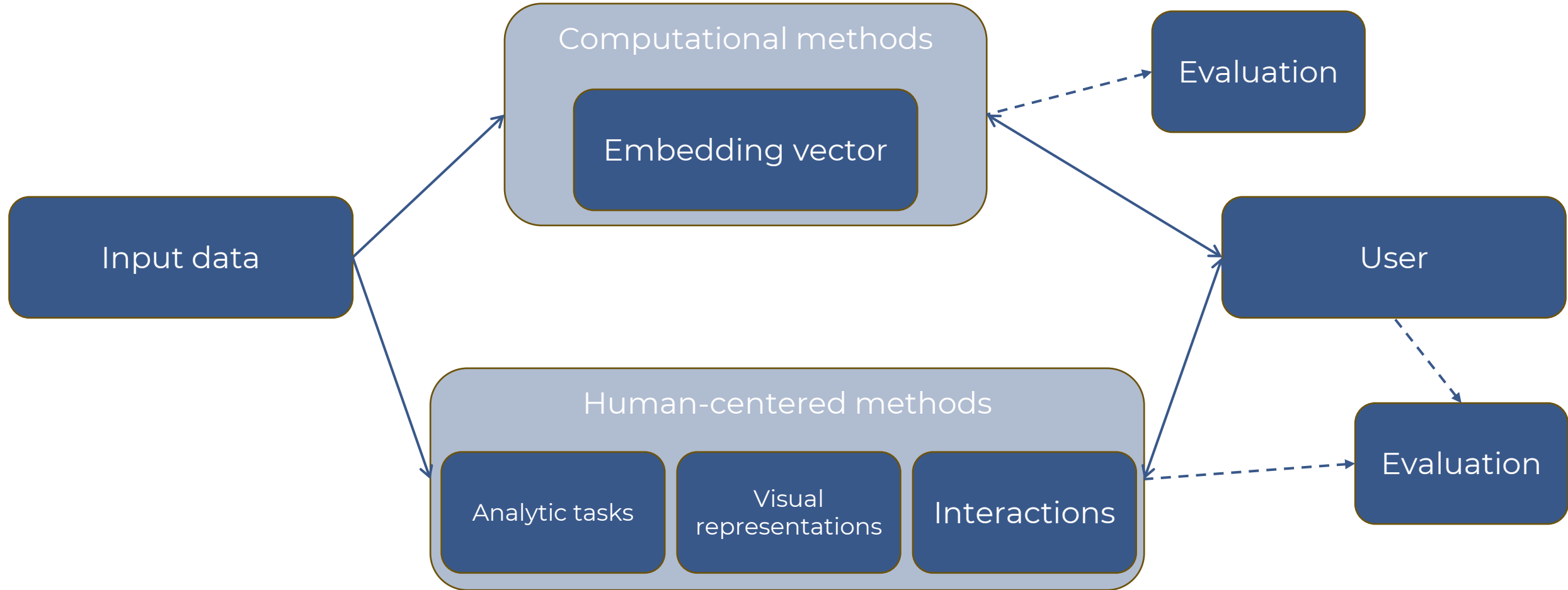
0.3 0.3 0.1 0.7 ... 0.2 0.1



Embedding
model

0.6 0.2 0.9 0.8 ... 0.9 0.5

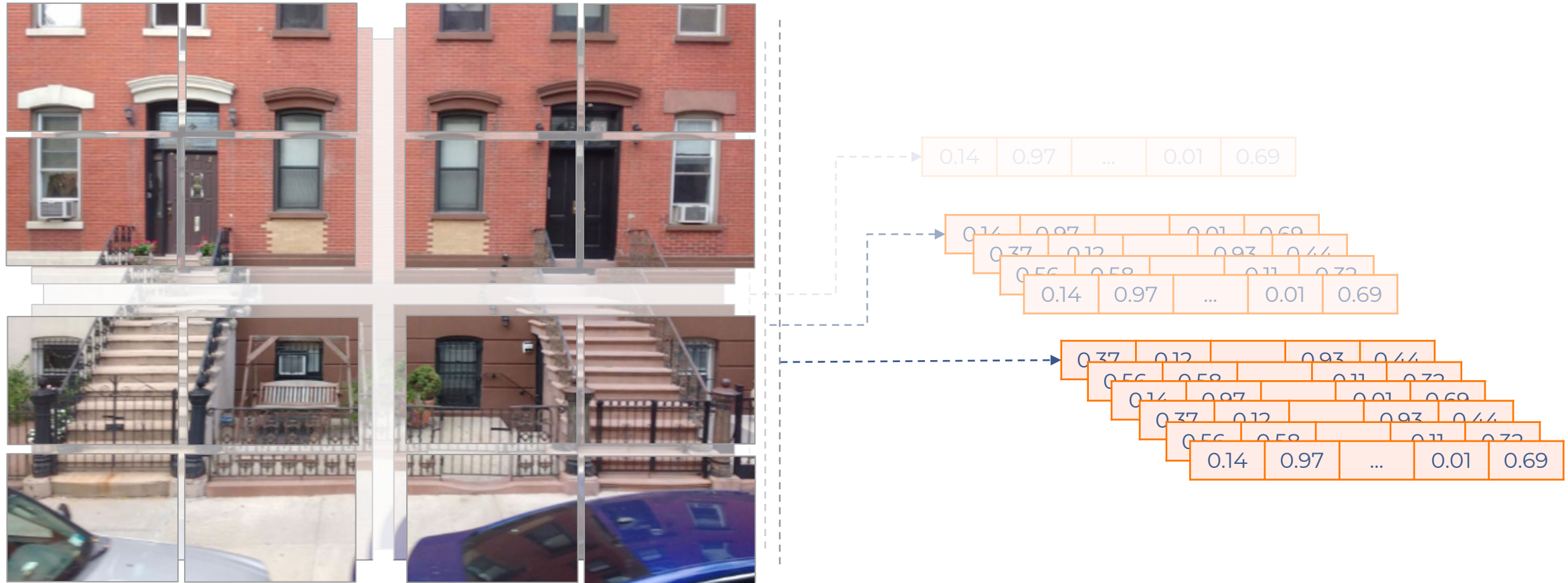
What are embeddings?



Human-in-the-loop processes can augment computer-centric components.

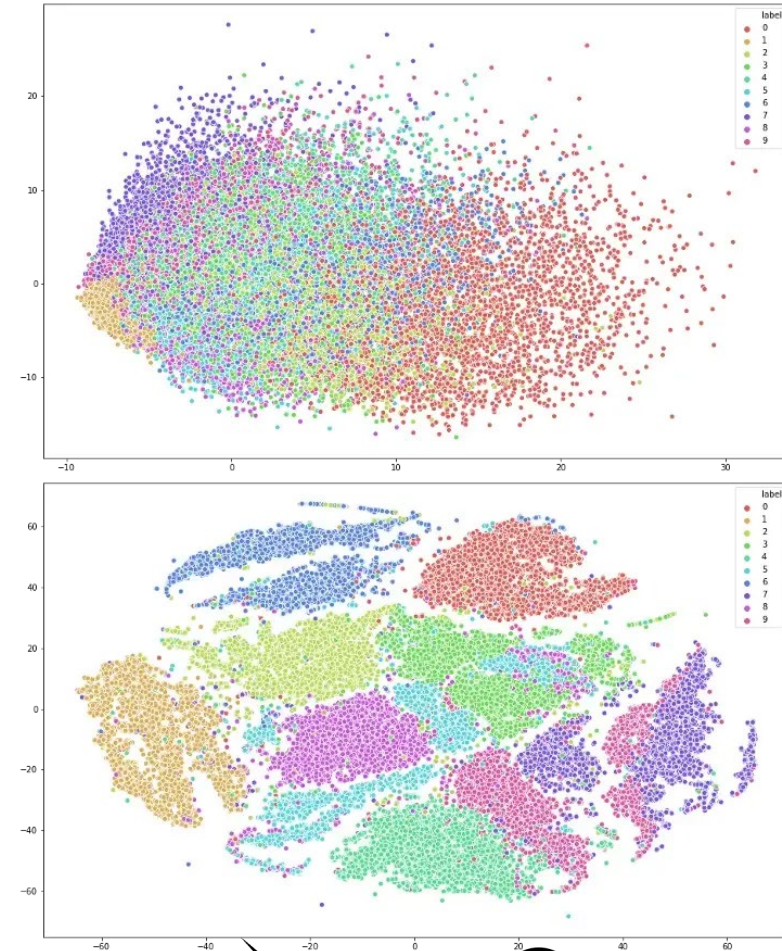
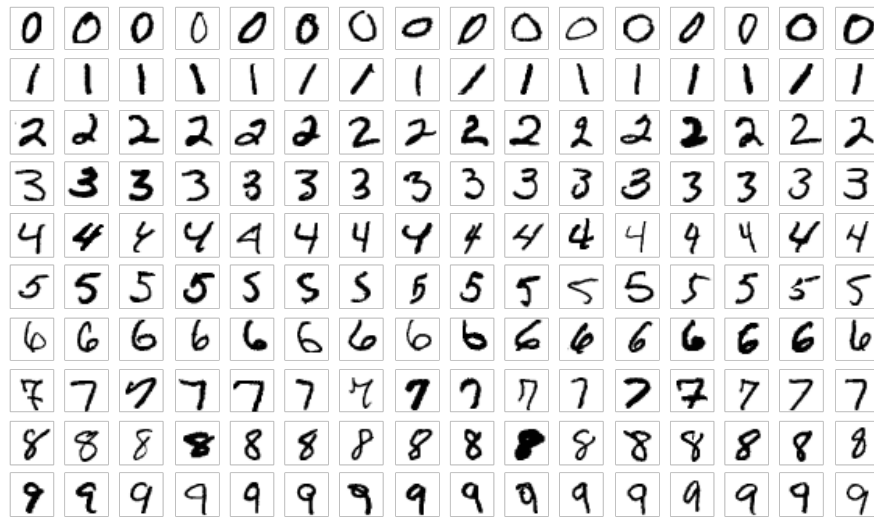


Image embeddings

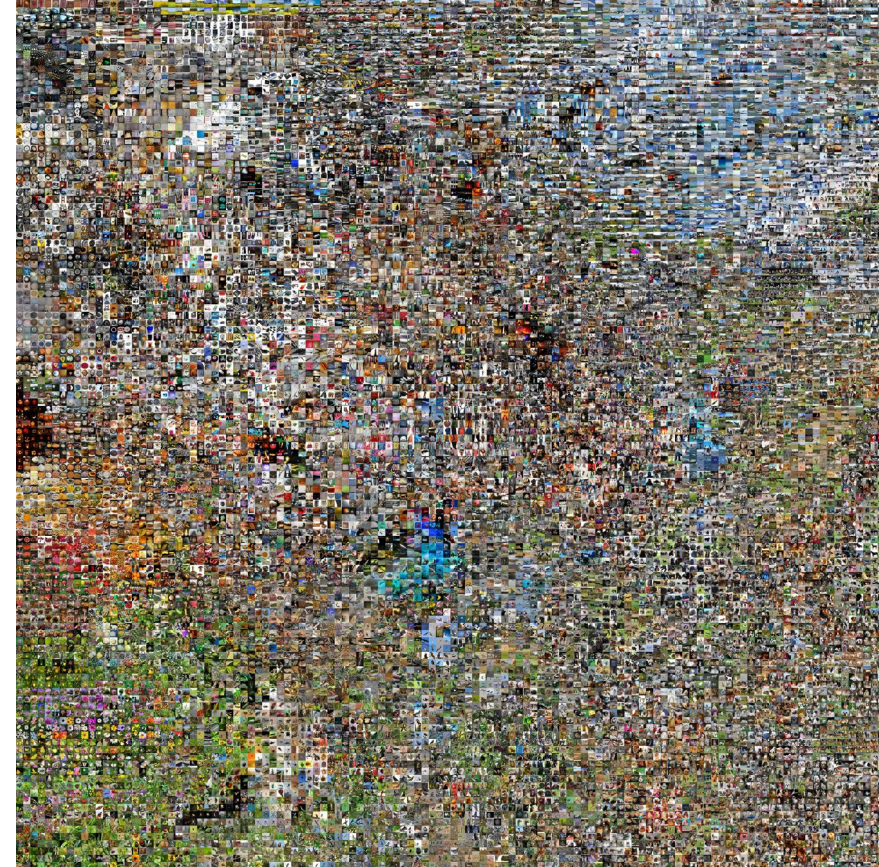


Technique	Taxonomy												
	Data Types					Linearity	Supervision	Multi-level	Locality	Steerability	Stability	OOC	Comp.
	Di	Or	Ca	Ne	Ct								
PCA [65]			✓			✓					•	✓	$O(p^3)$
LDA [50]			✓			✓	✓					✓	$O(np^2+p^3)$
Classical MDS [155]	✓		✓	•		✓	•			•	•		$O(n^3)$
Kruskal [79]	✓	✓	✓	•			•			•		•	$O(in^2)$
NLM [132]	✓		✓	•			•			•		•	$O(in^2)$
MCA [17]			•		✓								$O(n^3)$
Smacof [42]	✓	✓	✓	•			•					•	$O(in^2)$
SOM [126]			✓									✓	$O(l^2np+l^2)$
FastMap [48]	✓		✓	•			•					•	$O(n)$
Chalmers [32]	✓		✓	•			•	•		•	◦	✓	$O(in)$
GTM [22]			✓									•	$O((lnp)^3+m^3)$
Pekalska [120]	✓		✓	•		✓	•						$O(r^3+rn)$
CCA [43]	✓		✓	•			•					✓	$O(l^2)$
LLE [129]	✓		✓	•			•		✓			•	$O(n^3)$
Isomap [153]	✓		✓	✓			•					•	$O(n^3)$
Lapl. Eigenmaps [15]	✓		✓	✓			•		✓			•	$O(n^3)$
Force-Directed [152]	✓		✓				•	•				✓	$O(in^2)$
LTSA [180]			✓			•			✓				$O(n^3)$
MVU [169]	✓		✓	•			•		✓			•	$O(n^3)$
LSP [117]	✓		✓	✓			•	•	✓	✓	◦		$O(n^3)$
SNE [64]	✓		✓	•			•	•		◦		•	$O(in^2)$
PLMP [118]			✓			•	•	•	✓	✓	◦		$O(n^3)$
LAMP [73]			✓				•	•	✓	✓	✓	✓	$O(pn)$
RBF-MP [2]	✓		✓	•			•			✓	◦	•	$O(r^3+n)$
LoCH [47]	✓		✓	•			•		✓	✓		✓	$O(n\sqrt{(n)})$
ClassiMap [90]	✓		✓	•			✓					•	$O(in^2)$
Kelp [13]	✓		✓	•		•	•		✓	✓	◦	•	$O(r^3)$

- Dimensionality reduction



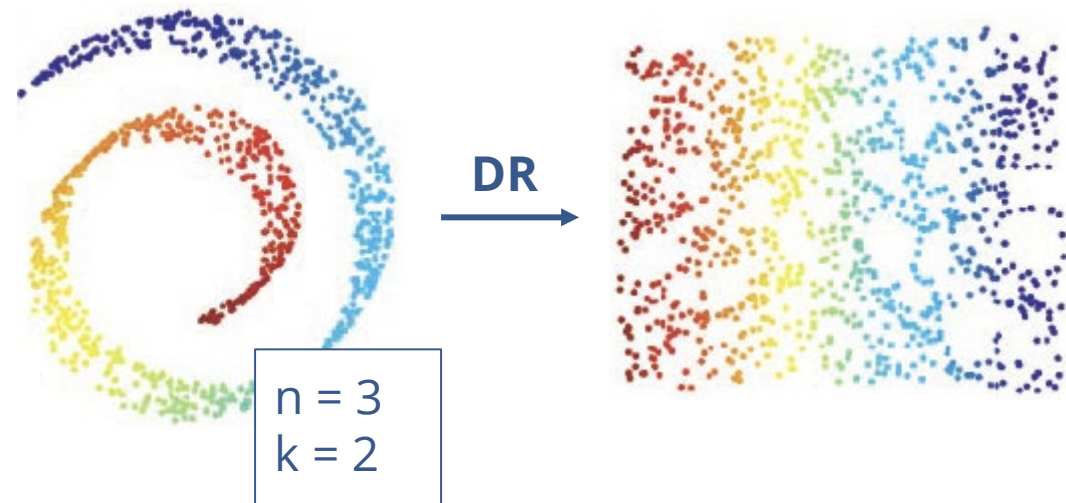
Dimensionality reduction



Dimensionality reduction

- High-dimensional data typically has an intrinsic dimension smaller than the space in which it is embedded.

DR comprises a category of unsupervised methods that aim to capture, to a certain extent, the intrinsic dimension of the data.

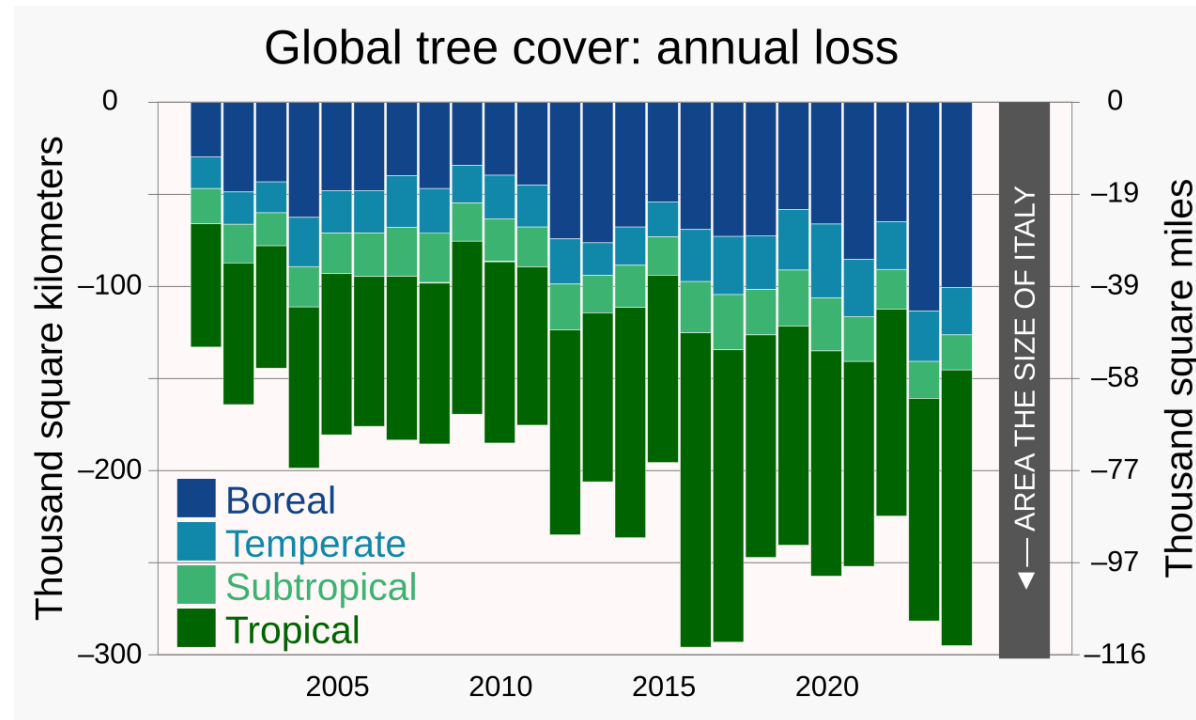


Visual analytics

- *“Visual analytics is the formation of abstract visual metaphors in combination with a human information interaction that enables detection of the expected and discovery of the unexpected within massive, dynamically changing information spaces.”*

[Wong and Thomas, 2004]

Going forward

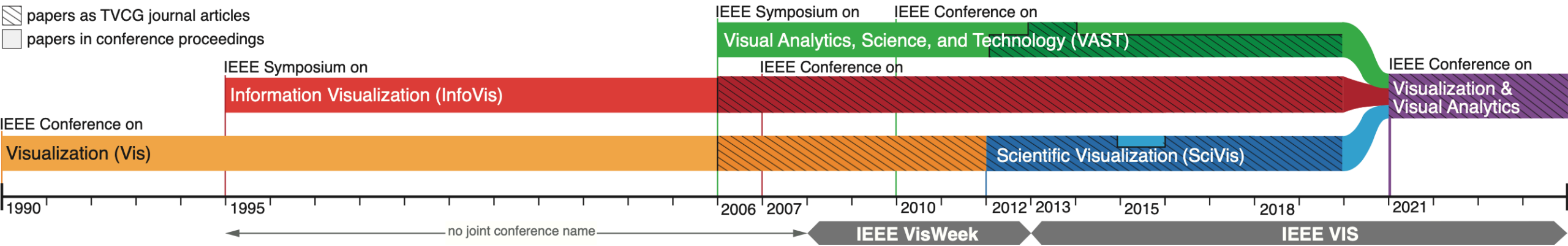


A bar chart is one sentence in the language of visualization.

Going forward



Going forward



Going forward

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DEPARTMENT: VISUAL COMPUTING: ORIGINS

On the Birth and Impact of the IEEE Visualization Conference

Lawrence J. Rosenblum, consultant, Rockville, MD, 20850, USA

IN THE BEGINNING

During the 1980s, I had been utilizing computer graphics and machine vision algorithms to extract knowledge from ocean science data-sets. The use of color, animation, volume rendering, and machine vision produced advances over topics spanning physical oceanography, ocean geophysics, underwater acoustics, and sonar design.¹ This was hardly unique, as similar efforts transpired across science and engineering, although progress was limited by the cost of even minimally capable hardware. Also, most scientists, engineers, and administrators had yet to comprehend the benefits that visualization could offer. "At most I'll get some pretty pictures" was the reply of a Naval Research Laboratory (NRL) Branch Head when I proposed spending time and funds to produce and analyze color images of "fine-structure" (small-scale variations in the ocean's physical properties, such as temperature, salinity, and density, which occur over vertical and horizontal scales ranging from a meter to several hundred meters). George Marmorino and I used color and image processing to demonstrate a longstanding conjecture about fine-structure behavior in a frontal region² and to identify the underlying physical processes.³ Colin Shen and I⁴ used animation to enhance understanding of a phenomenon called saltfingering. The Branch Head changed his tune.

Computers became smaller, cheaper, and more powerful, thus offering scientists and engineers previously unimagined capabilities. Applying visualization techniques to scientific data started to be recognized

as a valuable tool for understanding data. Those whose focus was on the development of visualization algorithms and toolsets began to organize meetings and symposia to present their work and learn about new methods from other practitioners. In short, a research field was emerging.

LIKE THE PHOENIX

The IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity. Today, it has over 460,000 members in more than 190 countries, with more than 66% from outside the United States. IEEE has 39 Technical Societies (TS) and eight Technical Councils, which serve to advance S&T in computer science and engineering. IEEE members are engineers, scientists, and allied professionals whose technical interests are rooted in electrical and computer sciences, engineering, and related disciplines. The IEEE Computer Society (CS) is the largest TS in the IEEE.

The CS performs standards development and educational activities, as well as organizes conferences where participants present their work, attend tutorials and workshops, and meet with colleagues both formally and ad hoc in the hallways. Many IEEE conferences are organized by a Technical Community (TC). Some TC's are large and active. However, the Computer Graphics TC (VGTC)⁵ was a small TC focused on computer graphics education in the early 1980s and was completely inactive in 1986.

The CS asked if I was interested in revitalizing the TC. I suspect that I was not the first person asked, as the Association for Computing Machinery (ACM) already had a Special Interest Group (SIG) on

COMES THE DAWN

After nearly three years of planning, the time had finally arrived for Visualization'90. Most start-up conferences have an established, formal base of practitioners from which to draw attendees. We didn't, adding a significant risk factor to our efforts. Well, we would soon have the answer to the question that haunted us: "Would anyone attend?" The day before the opening session was devoted to tutorials. The tutorial chair had arranged an exciting program spanning key visualization issues. The lecturers were recognized as leaders in their field:

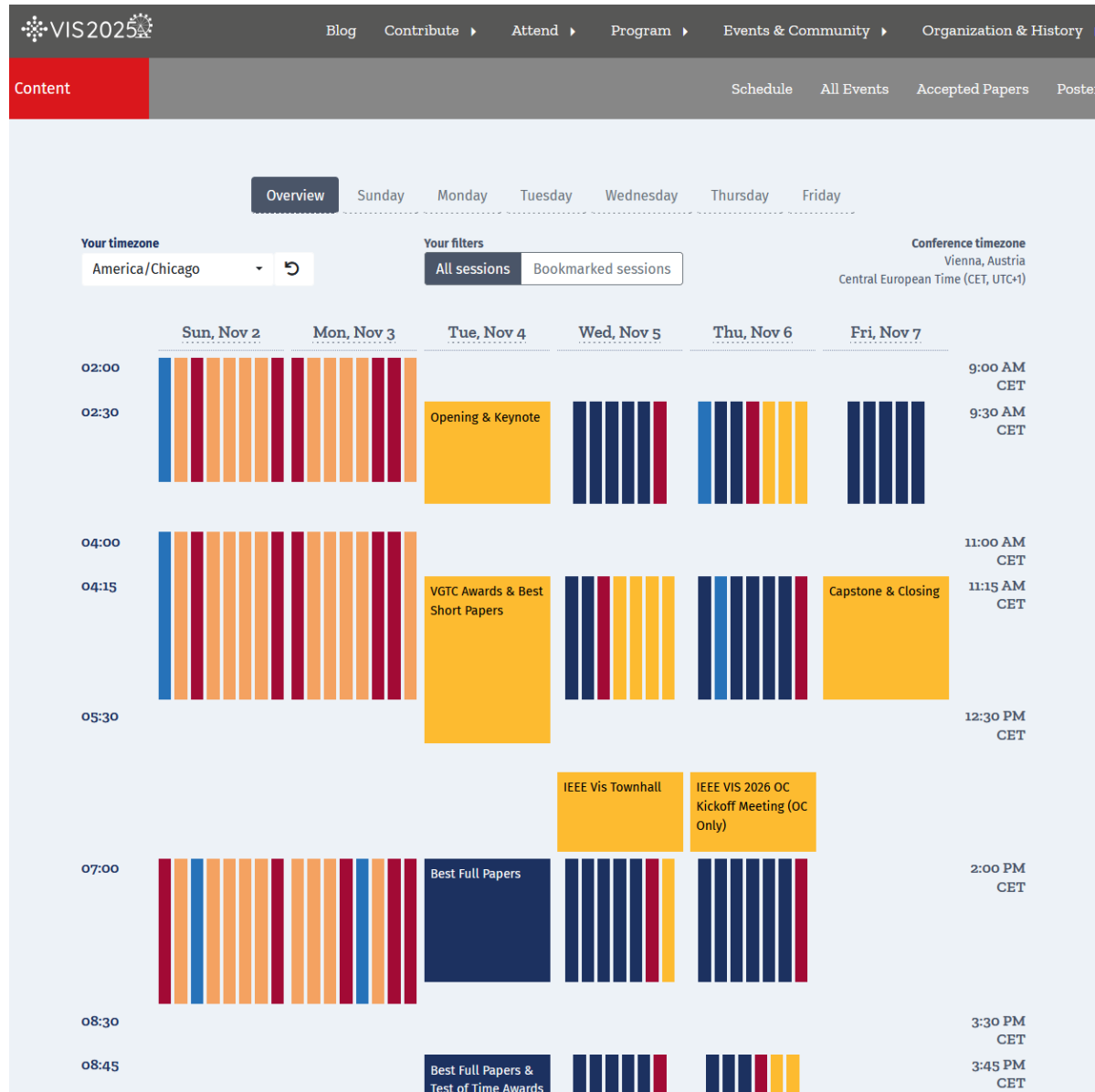
- Introductory and advanced computer vision: Azriel Rosenfeld.
- Volume visualization: Marc Levoy, Pat Hanrahan, Arie Kaufman, and William Lorenson.
- Visualizing multidimensional data: William Cleveland, Richard Becker, William Shye, and Allan Wilks.
- Knowledge visualization: Aaron Marcus.
- Visual programming environments: Ephraim Glinert.

The tutorials completed we spent a restless night awaiting the dawning of the new day portending the 9 A.M. Plenary Session with Gordon Bell as the Keynote



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Digital Object Identifier 10.1109/MCG.2025.3560905
Date of current version 18 July 2025.

⁵To emphasize visualization, TCCG changed its name to TCVG. Subsequently, it was changed to the Visualization and Graphics TC (VGTC) to distinguish it from the new IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS (TVCG). VGTC will be the name used throughout this article.





Grand Challenges in Bioinformatics Data Visualization

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Keywords: bioinformatics, data science, data visualization, visual analytics, computational biology, science communication

INTRODUCTION

Increasingly, the life sciences rely on data science, an emerging discipline in which visualization plays a critical role. Visualization is particularly important with challenging data from cutting-edge experimental techniques, such as 3D genomics, spatial transcriptomics, 3D proteomics, epiproteomics, high-throughput imaging, and metagenomics. Data visualization also plays an increasing role in how research is communicated. Some scientists still think of data visualization as optional; however, as more realize it is an essential tool for revealing insights buried in complex data, bioinformatics visualization is emerging as a subdiscipline. This article outlines current and future grand challenges in bioinformatics data visualization, and announces the first publication venue dedicated to this subdiscipline.

Over the past two decades, life science data have increased rapidly in volume and complexity, with the result that data analysis is often the major bottleneck (O'Donoghue et al., 2010a). For example, "All major genomics breakthroughs so far have been accompanied by the development of ground-breaking statistical and computational methods" (Green et al., 2020). Thus, in the remaining decades of the 21st century, life scientists will become increasingly reliant on the emerging tools and methods of data science (Blei and Smyth, 2017; Altman and Levitt, 2018).

One of these methods is data visualization (a.k.a. DataVis), which plays a critical role in transforming data and analysis outcomes into insight (Card et al., 1999). Data visualization involves analysis, design, and rendering, as well as observation and cognitive processing (Figure 1). Some scientists think of DataVis as an optional step mostly aimed at aesthetics — however, there is growing recognition that it is an essential tool in the analysis of complex data; two indicators of this recognition are the recent sales of DataVis companies Looker and Tableau for US\$3B and \$16B, respectively.

Currently, however, most attention is focused on another aspect of data science, namely, the use of machine learning to develop artificial intelligence systems. Such systems have recently led to exciting advances in the life sciences (e.g., Callaway, 2020a) — but also to some hyperbole. Clearly, machine learning methods are increasingly critical for research; but these methods also have limitations (Challen et al., 2019; Heaven, 2019; Yu and Kohane, 2019). More fundamentally, automated methods are insufficient, since analysis outcomes must be observed and understood by an analyst before insight can occur (Figure 1). Most analysts use data visualization as an integral part of their cognitive processes — especially important is manual validation, which involves checking for errors and outliers in raw data, and for wrong assumptions used in automated analysis methods (Anscombe, 1973).

Automated data analysis (including machine learning) and data visualization are just components of the larger goal of data science, which the eminent computer scientist Fred Brooks argues should focus on 'Intelligence Amplification' (a.k.a. I.A.) — i.e., on amplifying our abilities to manage more

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DEPARTMENT: VISUALIZATION VIEWPOINTS

Grand Challenges in Visual Analytics Applications

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In the past two decades, research in visual analytics (VA) applications has made tremendous progress, not just in terms of scientific contributions, but also in real-world impact across wide-ranging domains including bioinformatics, urban analytics, and explainable AI. Despite these success stories, questions on the rigor and value of VA application research have emerged as a grand challenge. This article outlines a research and development agenda for making VA application research more rigorous and impactful. We first analyze the characteristics of VA application research and explain how they cause the rigor and value problem. Next, we propose a research ecosystem for improving scientific value, and rigor and outline an agenda with 12 open challenges spanning four areas, including foundation, methodology, application, and community. We encourage discussions, debates, and innovative efforts toward more rigorous and impactful VA research.

Visual analytics (VA) applications leverage software artifacts with VA techniques to solve real-world problems. They combine automated data analysis techniques with interactive visualizations to facilitate effective reasoning and decision-making on large and complex datasets.¹ Since the establishment of the IEEE Conference on Visual Analytics Science and Technology in 2006, VA applications have grown into a significant and influential research field in visualization. For example, research on VA applications

occupies about 34% (41/120) of the full papers published in the IEEE Visualization Conference (IEEE VIS) 2022. VA applications have also achieved substantial societal impact by providing success stories on solving real-world problems, especially in wide-ranging high-impact domains, such as bioinformatics, urban analytics, explainable artificial intelligence (AI), and social media.

Nevertheless, decades of research and practices of VA applications have exposed multiple value- and rigor-related questions (see Figure 1). In light of these questions, we organized a panel at IEEE VIS 2022 to discuss grand challenges for making this research field more valuable and rigorous. The panel theme was informed by an informal interview study with 17 active researchers in VA system research.²

Grand Challenges in Immersive Analytics

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ABSTRACT

Immersive Analytics is a quickly evolving field that unites several areas such as visualisation, immersive environments, and human-computer interaction to support human data analysis with emerging technologies. This research has thrived over the past years with

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CHI '21, May 8–13, 2021, Yokohama, Japan
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ACM ISBN 978-1-4503-3996-4/21/05...\$15.00
<https://doi.org/10.1145/3411764.3416666>

multiple workshops, seminars, and a growing body of publications, spanning several conferences. Given the rapid advancement of interaction technologies and novel application domains, this paper aims toward a broader research agenda to enable widespread adoption. We present 17 key research challenges developed over multiple sessions by a diverse group of 24 international experts, initiated from a virtual scientific workshop at ACM CHI 2020. These challenges aim to coordinate future work by providing a systematic roadmap of current directions and impending hurdles to facilitate productive and effective applications for Immersive Analytics.



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FIGURE 1. Questions on the rigor and value of VA application research have emerged as a grand challenge.

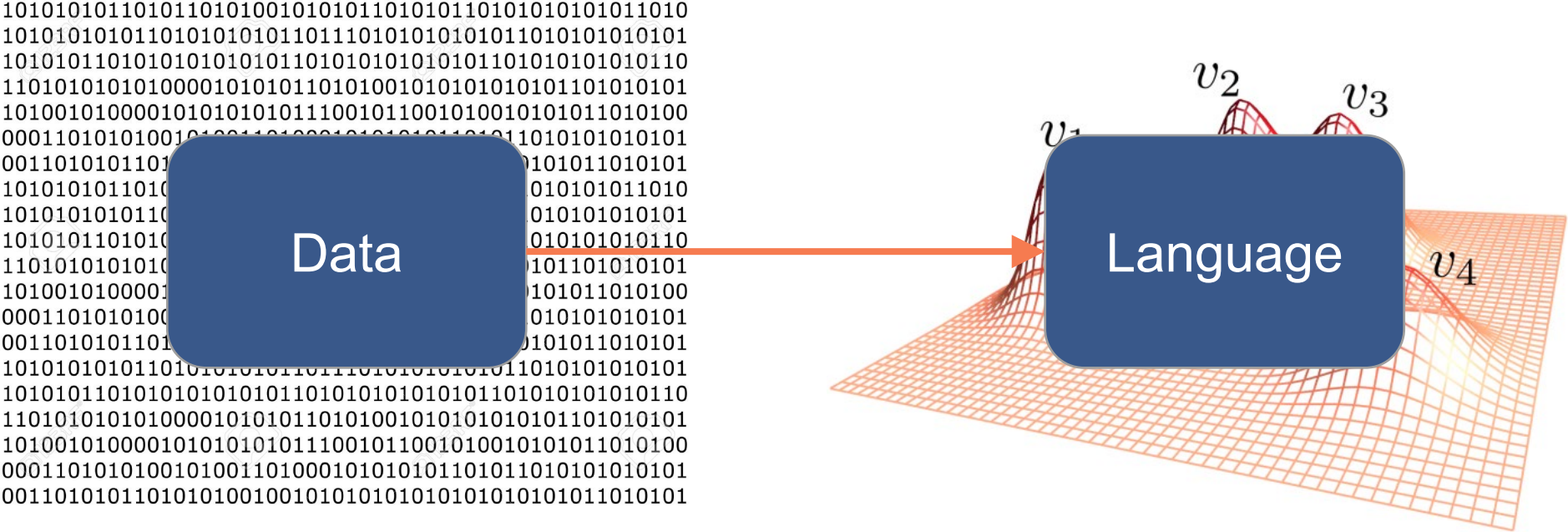
Going forward

Grand challenges in visualization

- How to build shared vocabularies and theories?
- How to provide guidance in complex VA systems?
- How can visualization and AI collaborate so that systems are powerful and interpretable?
- How do we build visualizations that are adaptive to people (their goals, expertise, accessibility needs, cognitive differences) without becoming opaque?



Topological data analysis



Topological data analysis

Taxi patterns: finding interesting events on spatiotemporal data

“Using Topological Analysis to Support Event-Guided Exploration in Urban Data” [TVCG 2014]

Urban Pulse: characterizing urban activity with spatiotemporal data

“Urban Pulse: Capturing the Rhythm of Cities” [TVCG 2016]

Using Topological Analysis to Support Event-Guided Exploration in Urban Data

Harish Doraiswamy Member, IEEE, Nivan Ferreira Student Member, IEEE, Theodoros Damoulas, Juliana Freire Member, IEEE and Cláudio T. Silva Fellow, IEEE

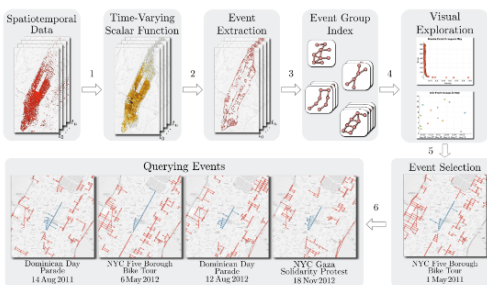


Fig. 1. Overview of the event-guided exploration technique. First, (1) the input data is transformed into a time-varying scalar function. (2) Topological features are computed from the scalar functions to identify the set of events. (3) An event group index is then created from the identified events to support efficient querying over a large number of events. (4) A visual interface guides the user towards interesting events (5) in the data, allowing them to select an event and (6) interactively search for similar events.

Abstract—The explosion in the volume of data about urban environments has opened up opportunities to inform both policy and administration and thereby help governments improve the lives of their citizens, increase the efficiency of public services, and reduce the environmental harms of development. However, cities are complex systems and exploring the data they generate is challenging. The interaction between the various components in a city creates complex dynamics where interesting facts occur at multiple scales, requiring users to inspect a large number of data slices over time and space. Manual exploration of these slices is ineffective, time consuming, and in many cases impractical. In this paper, we propose a technique that supports event-guided exploration of large, spatio-temporal urban data. We model the data as time-varying scalar functions and use computational topology to automatically identify events in different data slices. To handle a potentially large number of events, we develop an algorithm to group and index them, thus allowing users to interactively explore and query event patterns on the fly. A visual exploration interface helps guide users towards data slices that display interesting events and trends. We demonstrate the effectiveness of our technique on two different data sets from New York City (NYC): data about taxi trips and subway service. We also report on the feedback we received from analysts at different NYC agencies.

Index Terms—Computational topology, event detection, spatio-temporal index, urban data, visual exploration.

1 INTRODUCTION

Recent technological innovations have enabled the collection of enormous amounts of data pertaining to cities, from conventional sensors, such as power consumption [30] and noise [56], to more “unconventional” means of capturing city dynamics such as GPS in vehi-

cles [5, 22, 54], mobile devices [26], and social media [12, 29]. Cities all over the world are not only collecting these data, but they are also making the data available (see e.g., [43, 44, 45]). If properly analyzed, urban data can take us beyond today’s imperfect and often anecdotal understanding of cities to enable better operations, informed planning, and improved policy. These data are also a rich source for social scientists who aim to better understand cities and their populations.

However, there are many challenges involved in enabling the effective analysis of urban data. They stem not only from the volume of data, but also from the inherent spatio-temporal complexity of the underlying processes in a city. The usual approach to analyze this kind of data is to use different types of aggregation and produce visual summaries [2, 35]. These lead to a trade-off between the level of aggrega-

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Urban Pulse: Capturing the Rhythm of Cities

Fabio Miranda, Harish Doraiswamy, Member, IEEE, Marcos Lage, Kai Zhao, Bruno Gonçalves, Luc Wilson, Mondrian Hsieh, and Cláudio T. Silva, Fellow, IEEE

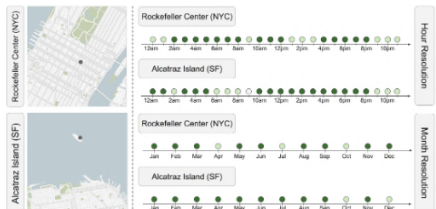


Fig. 1. Comparing two popular tourist locations – Rockefeller Center in New York City (NYC) and Alcatraz Island in San Francisco (SF), using the pulse of these locations. The pulse, defined by a set of beats over multiple resolutions, captures the level of activity at a given location. In this example, the beats for the hourly and monthly resolutions are shown, based on Flickr activity. They are computed based on the topology of the time-varying scalar function that models the spatio-temporal distribution of the activity corresponding to a city. Dark green represents a significantly high activity at the location, while light green represents a relatively high activity compared to its neighboring locations. The similarity between the pulses of the two locations over the different resolution indicates that the level of activity is similar across time steps and resolutions even though one is located on the mainland, while the other is an island.

Abstract—Cities are inherently dynamic. Interesting patterns of behavior typically manifest at several key areas of a city over multiple temporal resolutions. Studying these patterns can greatly help a variety of experts ranging from city planners and architects to human behavioral experts. Recent technological innovations have enabled the collection of enormous amounts of data that can help in these studies. However, techniques using these data sets typically focus on understanding the data in the context of the city, thus failing to capture the dynamic aspects of the city. The goal of this work is to instead understand the city in the context of multiple urban data sets. To do so, we define the concept of an “urban pulse” which captures the spatio-temporal activity in a city across multiple temporal resolutions. The prominent pulses in a city are obtained using the topology of the data sets, and are characterized as a set of beats. The beats are then used to analyze and compare different pulses. We also design a visual exploration framework that allows users to explore the pulses within and across multiple cities under different conditions. Finally, we present three case studies carried out by experts from two different domains that demonstrate the utility of our framework.

Index Terms—Topology-based techniques, urban data, visual exploration.

1 INTRODUCTION

Cities have been the loci of economic activity and human development through the ages, and will continue to be the source of many of the innovations and solutions to the challenges faced by society. In less than a century, the world has moved from less than 20% of the population residing in urban areas to over 50%. A number that is expected to rise to 70% by 2050, and up to 90% in North America alone [44]. This rapid increase in urbanization gives rise to several challenges on how

to efficiently transport, house, educate, employ and even entertain an ever increasing number of citizens on a daily basis. A better understanding of the structure of the population, and on how it is distributed and evolving can greatly help in this endeavor.

Technological innovations have enabled the automatic collection of a diverse set of data about our daily lives both as individuals and as a society. As a consequence, cities are not only collecting, but also making unique data sets available through open data portals and live feeds [21, 30, 42]. An example that has quickly been followed by several private firms [31, 32] that understand the power of open data and of the potential that can be unlocked by combining diverse sources of data. This opens up new opportunities for city governments and social scientists to engage in data-driven science to better understand cities, and improve the lives of their residents. Not surprisingly, there have been several techniques proposed in recent times with the above goal in mind (e.g. see [2, 3, 13, 19, 20, 36]). However, all of these techniques focus on identifying features or events in the data, or exploring the mobility of predefined entities. That is, the focus in these works was on the data in the context of a city.

Our goal in this work is to understand the city in the context of



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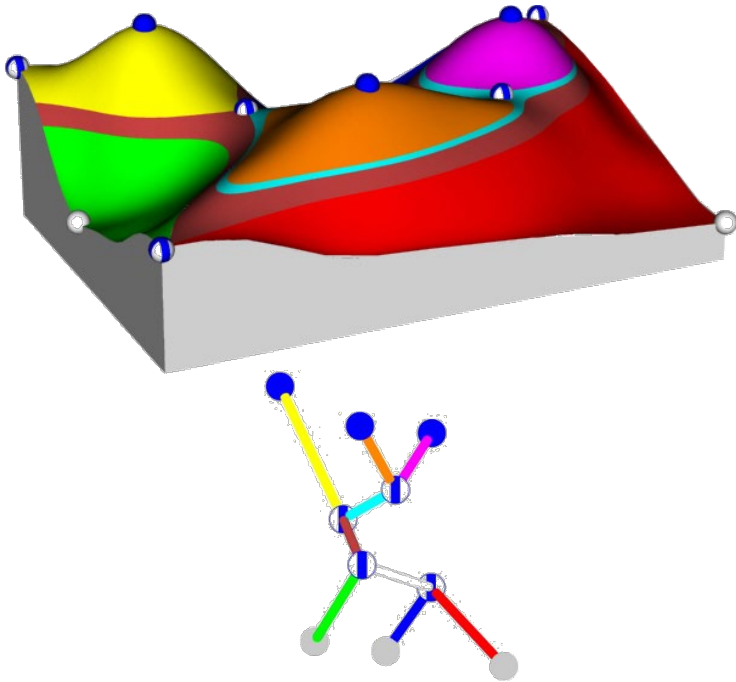
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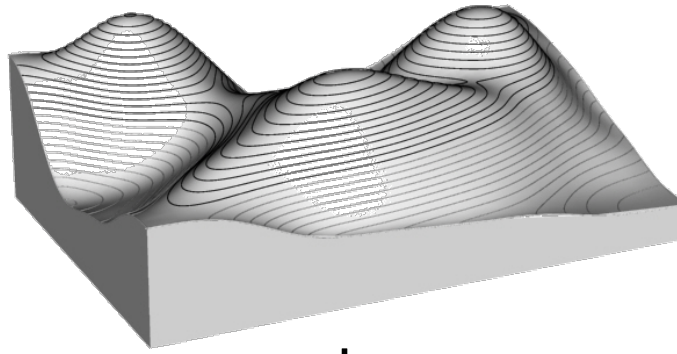
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Topological representations

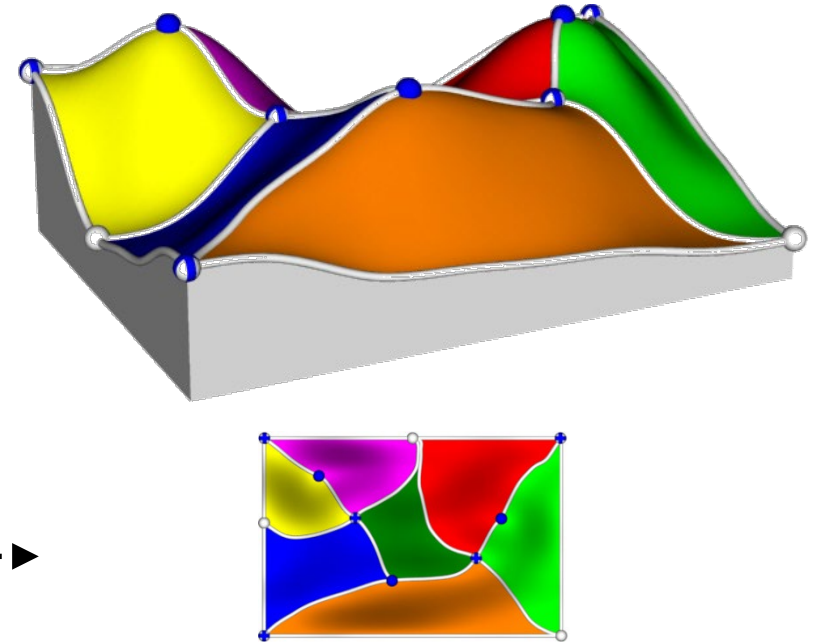
Contour trees
Reeb graphs
Merge trees



Scalar function



Morse-Smale complexes



← ———— | ———— →
[Gyulassy, VIS 2018]

A Comparison of Spatiotemporal Visualizations for 3D Urban Analytics

Roberta Mota, Nivan Ferreira, Julio Daniel Silva, Marius Horga, Marcos Lage, Luis Ceferino, Usman Alim, Ehud Sharlin, Fabio Miranda



Fig. 1. The four spatiotemporal visualizations compared in our study: (a) Embedded View, (b) Linked View, (c) Temporal Juxtaposition, and (d) Spatial Juxtaposition. In this example, the visualizations encode the values of an artificially-created temporal attribute of the building facade with four time steps. In each visualization, three rectangular regions with colored outlines are placed along the building surface. When comparing their attribute values, the blue region has the minimum average attribute value between t_1, t_2 time steps, whereas the red and black regions have the maximum average values in t_3 and t_4 , respectively.

Abstract—Recent technological innovations have led to an increase in the availability of 3D urban data, such as shadow, noise, solar potential, and earthquake simulations. These spatiotemporal datasets create opportunities for new visualizations to engage experts from different domains to study the dynamic behavior of urban spaces in this under explored dimension. However, designing 3D spatiotemporal urban visualizations is challenging, as it requires visual strategies to support analysis of time-varying data referent to the city geometry. Although different visual strategies have been used in 3D urban visual analytics, the question of how effective these visual designs are at supporting spatiotemporal analysis on building surfaces remains open. To investigate this, in this paper we first contribute a series of analytical tasks elicited after interviews with practitioners from three urban domains. We also contribute a quantitative user study comparing the effectiveness of four representative visual designs used to visualize 3D spatiotemporal urban data: spatial juxtaposition, temporal juxtaposition, linked view, and embedded view. Participants performed a series of tasks that required them to identify extreme values on building surfaces over time. Tasks varied in granularity for both space and time dimensions. Our results demonstrate that participants were more accurate using plot-based visualizations (linked view, embedded view) but faster using color-coded visualizations (spatial juxtaposition, temporal juxtaposition). Our results also show that, with increasing task complexity, plot-based visualizations perform better in preserving efficiency (time, accuracy) compared to color-coded visualizations. Based on our findings, we present a set of takeaways with design recommendations for 3D spatiotemporal urban visualizations for researchers and practitioners. Lastly, we report on a series of interviews with four practitioners, and their feedback and suggestions for further work on the visualizations to support 3D spatiotemporal urban data analysis.

Index Terms—Visualization, urban analytics, urban data, spatiotemporal data, empirical evaluation.

1 INTRODUCTION

Cities are the loci of economic activity and attract people looking for the myriad of services only offered by urban centers. With the growth

of urbanization, the process and problems that shape and form cities become far more intertwined—transportation, housing, street layouts, and land use are all becoming more complex and interconnected by the day. In their attempt to meet the demands of an increasing number of urban dwellers, cities resort to planning initiatives that lead to the verticalization of their landscape. Far from being just another element of cities, their verticality is perhaps the key feature that defines cities. While serving as a potential solution to many of the problems faced by cities (e.g., housing [8, 9]) verticalization can create new problems or exacerbate existing ones, e.g., their impact on overall temperature [38]. Cognizant of this phenomenon and the fact that cities are characterized by their verticality, a growing number of domains (e.g., civil engineering, urban planning, architecture) have gone beyond the usual flatland that defines a spatial region, and incorporated the third dimension (and its properties) in many of their analysis tasks in order to study the dynamic behavior of urban spaces. These tasks often rely on data that is intrinsic to the surface of buildings. Some examples include sound propagation simulation for studying noise in the urban environment [74], solar energy potential on facades for the installation of renewable energy equipment [41], outdoor heat exposure [49], sunlight access for the design of living spaces [37, 53, 55], and cellphone signal propagation

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Manuscript received 31 March 2022; revised 1 July 2022; accepted 8 August 2022.
Date of publication 27 September 2022; date of current version 2 December 2022.
This article has supplementary downloadable material available at <https://doi.org/10.1109/TVCG.2022.3209474>, provided by the authors.
Digital Object Identifier no. 10.1109/TVCG.2022.3209474

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DESIGN RECOMMENDATIONS

ALL TASKS

TIME

SpatialJX, TemporalJX

ERROR

LinkedV, EmbeddedV

TASK COMPLEXITY CONDITIONS

TIME

No. spatial instances

LinkedV, EmbeddedV

ERROR

No. spatial instances

LinkedV, EmbeddedV

No. time instances

LinkedV, EmbeddedV, SpatialJX

Camera navigation

LinkedV

View angle

LinkedV, EmbeddedV, SpatialJX, TemporalJX

ARGUS: Visualization of AI-Assisted Task Guidance in AR

Sonia Castelo*, Joao Rulff*, Erin McGowan*, Bea Steers, Guande Wu, Shaoyu Chen, Iran Roman, Roque Lopez, Ethan Brewer, Chen Zhao, Jing Qian, Kyunghyun Cho, He He, Qi Sun, Huy Vo, Juan Bello, Michael Krone, and Claudio Silva

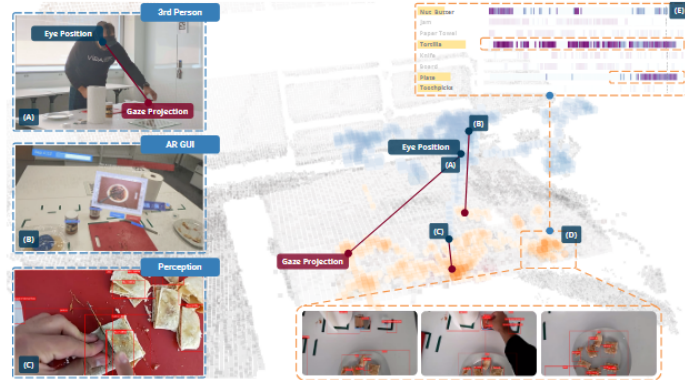


Fig. 1: ARGUS is a visual analytics tool for real-time and historical evaluation of sensor and model outputs of AR task assistants. Shown here are (A) 3rd-person perspective of a human (performer) performing a task with AR headset guidance, (B) the AR GUI from the perspective of the performer, (C) a snapshot of a visual perception model analyzing data from the headset camera. For each time step, detailed information highlights the performer's gaze direction and the corresponding frame recorded by the headset. (D) heatmaps of the performer's gaze projection onto the world point cloud allow for inspection and understanding of their attention over time. The cluster on the left shows the performer finalizing a recipe. (E) object detection information from ARGUS Temporal View illustrating that *Plate* was only detected towards the end of the task while *Tortilla* was detected throughout the whole session.

Abstract—The concept of augmented reality (AR) assistants has captured the human imagination for decades, becoming a staple of modern science fiction. To pursue this goal, it is necessary to develop artificial intelligence (AI)-based methods that simultaneously perceive the 3D environment, reason about physical tasks, and model the performer, all in real-time. Within this framework, a wide variety of sensors are needed to generate data across different modalities, such as audio, video, depth, speech, and time-of-flight. The required sensors are typically part of the AR headset, providing performer sensing and interaction through visual, audio, and haptic feedback. AI assistants not only record the performer as they perform activities, but also require machine learning (ML) models to understand and assist the performer as they interact with the physical world. Therefore, developing such assistants is a challenging task. We propose ARGUS, a visual analytics system to support the development of intelligent AR assistants. Our system was designed as part of a multi-year-long collaboration between visualization researchers and ML and AR experts. This co-design process has led to advances in the visualization of ML in AR. Our system allows for online visualization of object, action, and step detection as well as offline analysis of previously recorded AR sessions. It visualizes not only the multimodal sensor data streams but also the output of the ML models. This allows developers to gain insights into the performer activities as well as the ML models, helping them troubleshoot, improve, and fine-tune the components of the AR assistant.

Index Terms—Data Models; Image and Video Data; Temporal Data; Application Motivated Visualization; AR/VR/Immersive.

1 INTRODUCTION

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Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxxx

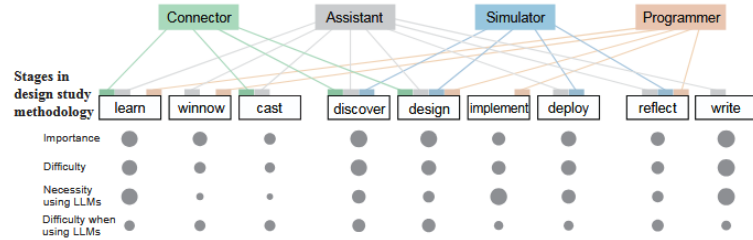
The concept of an augmented reality (AR) assistant has captured the human imagination for years, becoming a staple of modern science fiction through popular franchises such as the *Marvel Cinematic Universe*, *Star Trek*, and *Terminator*. The applications of such a system are seemingly endless. Humans, even those with domain expertise, are fallible creatures with imperfect memories whose skills deteriorate over time, especially during repetitive tasks or under stress. An AR assistant could help experts and novices alike in performing both familiar and new tasks. For instance, an AR assistant could aid a surgeon performing a



Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles

Shaolun Ruan , Rui Sheng , Xiaolin Wen , Jiachen Wang , Tianyi Zhang ,
Yong Wang , Tim Dwyer , and Jiannan Li 

Roles for LLM-assisted design study



Ratings for researchers' attitudes toward with and without LLMs (1 point = ~ 7 points)

Fig. 1: An overview of the summarized roles of LLMs in the LLM-assisted design study process. Following the framework proposed by Sedlmair et al. [34], we coded all nine stages of the design study by the compiled roles: 1) *Connector*: LLMs are used to bridge gaps between visualization researchers and domain experts; 2) *Assistant*: LLMs are commonly used for repetitive tasks to enhance productivity; 3) *Simulator*: LLMs support simulating user behavior, predicting feedback, and testing workflows to uncover usability issues and hidden use cases; 4) *Programmer*: LLMs are asked to assist in coding tasks, from prototyping to debugging. The bottom section shows ratings from 30 researchers on the importance, difficulty, and necessity of using LLMs, as well as the difficulty when using LLMs at each stage.

Abstract—Design studies aim to develop visualization solutions for real-world problems across various application domains. Recently, the emergence of large language models (LLMs) has introduced new opportunities to enhance the design study process, providing capabilities such as creative problem-solving, data handling, and insightful analysis. However, despite their growing popularity, there remains a lack of systematic understanding of how LLMs can effectively assist researchers in visualization-specific design studies. In this paper, we conducted a multi-stage qualitative study to fill this gap, which involved 30 design study researchers from diverse backgrounds and expertise levels. Through in-depth interviews and carefully-designed questionnaires, we investigated strategies for utilizing LLMs, the challenges encountered, and the practices used to overcome them. We further compiled the roles that LLMs can play across different stages of the design study process. Our findings highlight practical implications to inform visualization practitioners, and also provide a framework for leveraging LLMs to facilitate the design study process in visualization research.

Index Terms—Design Study, Large Language Models (LLMs), Qualitative Study, Visualization

1 INTRODUCTION

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Manuscript received xx.xxx.201x; accepted xx.xxx.201x. Date of Publication xx.xxx.201x; date of current version xx.xxx.201x. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxx

Sedlmair et al. introduced the term *design study* to describe an applied research methodology that focuses on creating visualizations to address specific, real-world problems [34]. More specifically, they defined a visualization design study as “a project in which visualization researchers analyze a specific problem faced by domain experts and design a visualization system that supports solving this problem” [34]. Building upon this definition, a nine-stage methodology framework was proposed for conducting design studies, as shown in Fig. 1, which has become a common guidance for this field of research in the visualization community. Design studies are crucial because they bridge the gap between theoretical visualization research and practical applications, ensuring that visualization tools are both usable and useful in real-world contexts [29]. Basically, researchers conduct design studies through labor-intensive processes (e.g., connecting with domain experts, extracting requirements via interviews, and iteratively designing and developing visualization systems), which require significant effort and refined knowledge to gain effective solutions.

Meanwhile, Large Language Models (LLMs) have shown increasing

A		B		C		D		
Ongoing degree	Vis expertise	Design study domain	# of research involved (accepted)	Mostly-used LLMs	Usage frequency of LLMs	Residence	Age	Self-described gender
Undergraduate	Novice	Real Estates	1 (0)	GPT-4o	11-20 ▀▀	Singapore	21-25	Male
Undergraduate	Novice	Education	1 (0)	GPT-4o	1-10 ▀	Singapore	21-25	Male
Undergraduate	Novice	Music	1 (0)	GPT-4o, GPT-o1	1-10 ▀	Singapore	16-20	Female
Undergraduate	Novice	Music	1 (0)	GPT-4o	11-20 ▀▀	Singapore	21-25	Female
Undergraduate	Novice	Real Estates	1 (0)	GPT-4o	21-30 ▀▀▀	Singapore	16-20	Female
Undergraduate	Novice	Real Estates	1 (0)	GPT-4o	21-30 ▀▀▀	Singapore	21-25	Male
Undergraduate	Novice	Healthcare	1 (0)	GPT-4o, GPT-o1	11-20 ▀▀	Singapore	21-25	Male
Undergraduate	Advanced beginner	Human-AI Collaboration	3 (1)	GPT-4o, Deepseek	21-30 ▀▀▀	Mainland China	21-25	Male
PhD	Intermediate	Text Visualization	5 (3)	GPT-4o, Claude 3.5	> 30 ▀▀▀▀▀	Australia	26-30	Male
PhD	Proficient	Finance; Software	18 (10)	GPT-4	21-30 ▀▀▀	Singapore	26-30	Male
PhD	Proficient	Biomedical	21 (3)	GPT-3.5	21-30 ▀▀▀	Hong Kong	26-30	Male
PhD	Proficient	Bioinformatics	16 (9)	GPT-4o	21-30 ▀▀▀	Singapore	26-30	Male
PhD	Proficient	E-learning	16 (6)	GPT-4	21-30 ▀▀▀	Hong Kong	21-25	Male
PhD	Proficient	Creative Design	17 (10)	GPT-4o, DeepSeek	11-20 ▀▀	Hong Kong	26-30	Female
PhD	Proficient	Color	15 (9)	GPT-4o, DeepSeek	21-30 ▀▀▀	Mainland China	21-25	Female
PhD	Proficient	Cultural Heritage	4 (2)	GPT-4o, DeepSeek	> 30 ▀▀▀▀▀	Mainland China	26-30	Male
PhD	Proficient	Medical; Notebook	10 (4)	GPT-4o, GPT-4, DeepSeek	21-30 ▀▀▀	Mainland China	21-25	Male
PhD	Proficient	XAI	15 (10)	GPT-4o, other (Kimi)	> 30 ▀▀▀▀▀	Mainland China	31-35	Male
PhD	Advanced beginner	Human-AI Collaboration	3 (1)	DeepSeek	> 30 ▀▀▀▀▀	Hong Kong	26-30	Male
PhD	Advanced beginner	AR/VR	6 (3)	GPT-4	21-30 ▀▀▀	Hong Kong	26-30	Male
PostDoc	Proficient	Storytelling	15 (9)	GPT-4	1-10 ▀	Australia	21-25	Female
PostDoc	Proficient	Notebook	13 (8)	GPT-4o, GPT-o1	21-30 ▀▀▀	Hong Kong	26-30	Female
Professor	Expert	Sports	20 (15)	GPT-4o	1-10 ▀	Mainland China	26-30	Male
Professor	Intermediate	Drug; Biomedical	21 (16)	DeepSeek	11-20 ▀▀	Mainland China	26-30	Female
Professor	Expert	Smart City	40 (20)	GPT-4	11-20 ▀▀	Mainland China	31-35	Male
Professor	Expert	Data Video	20 (10)	GPT-4o, DeepSeek	11-20 ▀▀	Hong Kong	31-35	Female
Professor	Expert	Bioinformatics	8 (5)	GPT-4o	11-20 ▀▀	Australia	41-45	Female
Professor	Proficient	Quantum Computing	15 (8)	GPT-4o, Claude 3.5	21-30 ▀▀▀	United States	46-50	Male
Professor	Proficient	Material Science	12 (8)	DeepSeek	21-30 ▀▀▀	Mainland China	41-45	Male
Professor	Proficient	High-performance Comp.	11 (9)	GPT-4o	1-10 ▀	United States	41-45	Female



NLI4VolVis: Natural Language Interaction for Volume Visualization via LLM Multi-Agents and Editable 3D Gaussian Splatting

Kuangshi Ai, Kaiyuan Tang, and Chaoli Wang

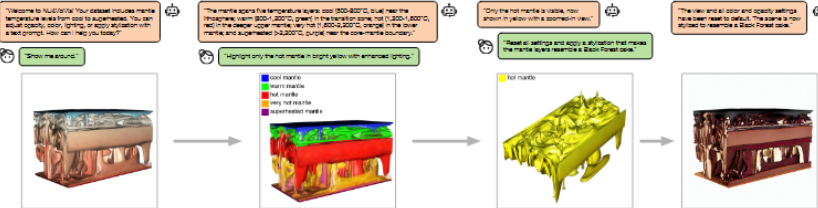


Fig. 1: An example of chatting with NLI4VolVis about the mantle temperature dataset. Our system lets users intuitively explore and edit volumetric datasets using natural language. Through open-vocabulary querying, real-time scene editing, and stylization, users can easily highlight semantic structures, adjust visualization properties, and apply creative styles.

Abstract—Traditional volume visualization (VolVis) methods, like direct volume rendering, suffer from rigid transfer function designs and high computational costs. Although novel view synthesis approaches enhance rendering efficiency, they require additional learning effort for non-experts and lack support for semantic-level interaction. To bridge this gap, we propose NLI4VolVis, an interactive system that enables users to explore, query, and edit volumetric scenes using natural language. NLI4VolVis integrates multi-view semantic segmentation and vision-language models to extract and understand semantic components in a scene. We introduce a multi-agent large language model architecture equipped with extensive function-calling tools to interpret user intents and execute visualization tasks. The agents leverage external tools and declarative VolVis commands to interact with the VolVis engine powered by 3D editable Gaussians, enabling open-vocabulary object querying, real-time scene editing, best-view selection, and 2D stylization. We validate our system through case studies and a user study, highlighting its improved accessibility and usability in volumetric data exploration. We strongly recommend readers check our case studies, demo video, and source code at <https://nli4volvis.github.io/>.

Index Terms—Volume visualization, novel view synthesis, natural language interaction, open-vocabulary querying, editable 3D Gaussian splatting, multi-agent collaboration

1 INTRODUCTION

In *scientific visualization* (SciVis), *volume visualization* (VolVis) enables domain experts to explore and analyze complex 3D volumetric datasets. Traditionally, *direct volume rendering* (DVR) has been the dominant approach, valued for its ability to produce high-quality visualizations via ray marching through dense voxel grids. However, DVR suffers from significant limitations: it is computationally intensive, requires substantial storage, and scales poorly with resolution, often rendering real-time interaction impractical.

To overcome these challenges, researchers have turned to *novel view synthesis* (NVS) techniques that generate new views from sparsely sampled rendering images, eliminating the need for full volumetric integration. Among these techniques, *3D Gaussian splatting* (3DGS) [10] has emerged as a promising alternative. 3DGS directly rasterizes point-based Gaussian primitives onto the screen, offering rapid rendering and compact scene representations well suited to modern GPUs. This approach enables more efficient and interactive visualization pipelines and serves as a foundation for flexible volumetric scene editing.

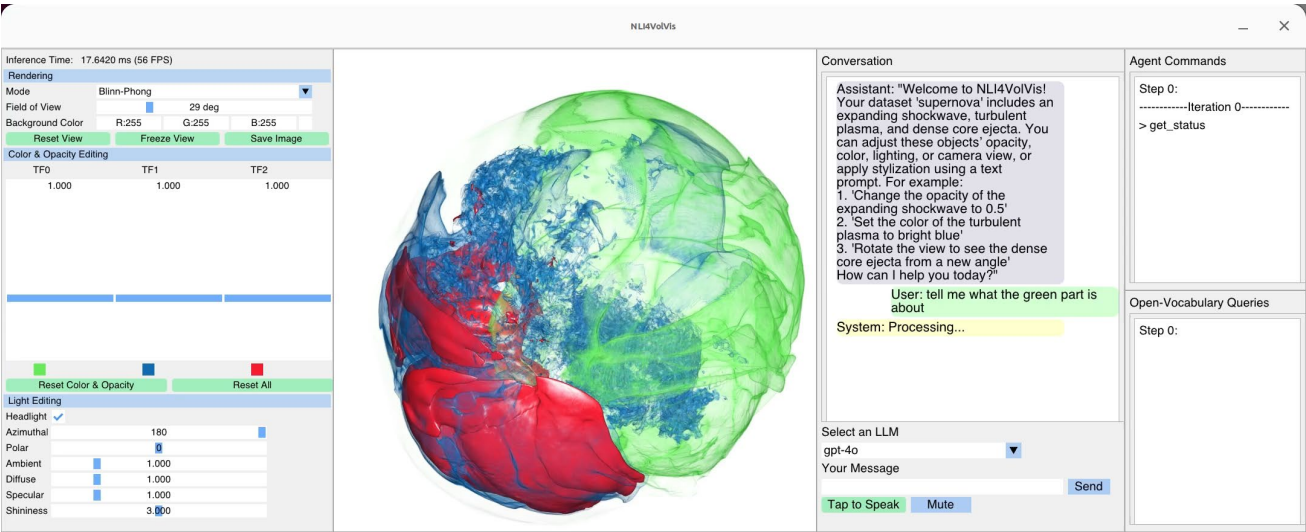
Despite such rendering improvements, the creation of meaningful visualizations still hinges on the *transfer function* (TF), which maps voxel intensities to color and opacity. While many advanced TF techniques have been proposed [4, 5, 9, 32, 34, 39] to enhance expressiveness, TF design remains a complex and often unintuitive task. Conventional TFs, in particular, lack semantic awareness, making object-level exploration and manipulation challenging in intricate datasets.

Recent advances in vision-language models have introduced automatic, text-driven TF generation [27]. However, these methods still fall short of enabling interactive, real-time scene editing through natural language, underscoring a persistent disconnect between *natural language interaction* (NLI) and intuitive, semantic-level control in VolVis.

The rise of *large language models* (LLMs) presents a transformative opportunity to bridge this gap. With strong reasoning and planning capabilities, LLMs can act as intelligent agents—interpreting user intent, orchestrating multi-step tasks, and generating context-aware responses. When combined with vision-language models, they facilitate semantic comprehension of visual data, enabling open-vocabulary interaction and editing. Moreover, *multi-agent systems*—comprising collaborative LLMs—can support distributed task execution, enhancing system robustness and flexibility.

Yet, no existing system has unified editable volumetric representations, open-vocabulary scene understanding, and collaborative multi-agent LLMs to support intuitive, natural language-based VolVis. Prior efforts [27, 29] either lack object-level semantic granularity or fall short of enabling real-time interaction and flexible scene manipulation.

To address these gaps, we present NLI4VolVis (Natural Language



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Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org.
Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxxx